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Refactoring Plan: Biostatistics Practicum

2026-09-07 14:05 PDT, revised 2026-09-07 17:40 PDT, merged with the readability plan, expert reviews, and LLM-section review 2026-10-09

Revision note

Map of this document. Findings 1 to 10 and Phases A to D are the September structural plan and its implementation record. Findings 11 to 23 come from the October reading pass, Finding 24 from a review of the LLM sections, Findings 25 to 30 from three expert reviews, and Finding 31 from a measurement of visual density with a list of 54 candidate tables and figures. Parts 3 to 7 hold the proposals; Part 6 is the consolidated defect register; Phases E to K are the remaining work; Part 11 records the author's decisions (resolved 2026-10-09; one minor item open).

2026-10-09 merge. This document now combines two plans. The first (2026-09-07) was a structural and adoptability audit based on a mechanical scan; Phases A to D below record what was implemented from it between 2026-09-07 and 2026-09-11 and what remains. The second (`READABILITY-PLAN.md`, 2026-10-09) came from a full line-by-line reading of all 47 source files plus a survey of learning-science and e-textbook guidance; it addressed prose, approachability, and memorability, which the first plan deliberately excluded. Its findings are Findings 11 to 23, and its proposals are Parts 3 to 7 and Phases E to J. `READABILITY-PLAN.md` has been removed; this file supersedes it.

2026-10-09, expert reviews and LLM sections. Added Finding 24 (a review of all 33 'Collaborating with an LLM' sections against current practice) with its specification in 4.6, and Findings 25 to 30 from three independent expert reviews (pharmaceutical biostatistics, academic biostatistics, industry data science), each of which recommended adoption only after major revisions. Their defects are consolidated in Part 6, whose new Tier 1 table defines Phase E1. Phase K (content gaps) and decisions 11 to 14 are new. Two earlier claims were strengthened rather than changed: the missing-data and SAP defects first noted in the reading pass were verified by computation, and the reviewers found them to be worse than first described.

2026-10-09, later. Added lessons from a structural reading of Hadley Wickham's *Advanced R* (2nd edition, adv-r.hadley.nz) and its companion *Advanced R Solutions*: section-level exercises, an Outline in the chapter card, margin footnotes for asides, nested numbered sections, a recurring visual grammar (4.5), short part introductions, a solutions companion, 'Getting help' and 'Meta-techniques' sections, and a generated colophon (5.5 to 5.9). Two of these reopen or extend decisions (Part 11, decisions 9 and 10).

Two positions taken by the September plan are reopened by the October reading rather than silently overridden. Part 9 ('What not to do') said not to rewrite prose for voice and not to split chapters on length. Both were listed as open decisions in Part 11 (since resolved), with the case for and against, because they reverse choices the author made deliberately.

2026-09-07 17:40 PDT. The first draft of this plan (14:05 PDT) contained three measurement errors, all corrected below and marked [**corrected 17:40 PDT**] at the point of correction. In summary: the license situation is materially worse than reported and is a four-way conflict rather than a single GPL-3 declaration; `11-quarto` has no computed figures rather than three; and the per-chapter answer sections, credited as a strength in the first draft, are placed at the maximum possible distance from the questions they answer, which is itself a named reviewer complaint. Two defects the first draft missed entirely have been added: the PDF ships with no table of contents, and the book contains no embedded media of any kind. Counts that changed on re-measurement are noted in Part 1.

Purpose and method

This document proposes a refactoring of *Biostatistics Practicum* for its primary audience: first-year master's students in data science or biostatistics, and entry-level data science engineers. It has two aims, pursued in sequence. The first is that the book be adoptable and practicable: legally open, navigable, with exercises that give feedback and code that runs. The second is that it be readable, approachable, and memorable for that audience.

Structural pass (2026-09-07 to 2026-09-08). Findings 1 to 10 rest on a mechanical scan of the 38 content files in `analysis/report/`, measured against the evidence base assembled for the blog post 'What Makes an Open Textbook Work' (`~/prj/ryyblog/posts/pub-open-textbook-guidebook/`). The proxies used were whether students can practice with feedback, whether the book is navigable and assignable in pieces, whether it is legally and practically adoptable, and whether it presents anything to look at. Counts are reproducible from the scripts in the appendix; where a number is an estimate, it says so.

Reading pass (2026-10-09). Findings 11 to 23 rest on three sources.

1. A full reading of all 47 `.qmd` files (about 136,000 words) in six groups, recording per-chapter metrics, openings, closings, prose samples, pedagogy, callout use, defects, and memorability devices, with `file:line` references.
2. A survey of about 30 searches and 8 page reads on learning science (Dunlosky et al. 2013; Mayer; Sweller; Ginns et al. 2013), technical-lesson design (Wilson, *Teaching Tech Together*; The Carpentries; Timbers and Cetinkaya-Rundel; R4DS; Happy Git; Missing Semester; Julia Evans), and on-screen reading (Nielsen Norman Group; Delgado et al. 2018).
3. Direct verification of a sample of the reading's claims.

Evidence labels follow the usual convention: **verified** (a command was run and the result observed), **inspected** (the source was read and the statement confirmed), and **inferred** (consistent with the pattern, not checked). Six reading-pass claims were independently re-checked and all held: the underpowered SAP (verified with `power.prop.test`), the daylight-saving example (inspected in `_freeze/`), the unused callout conventions (verified by `grep`), the merge-conflict example, the missing-data mechanism table, and the ADNI follow-up cutoff (all inspected).

Expert and specialist reviews (2026-10-09). Findings 24 to 30 rest on four further sources: a review of every LLM section against current practice (statements about current tools are expert knowledge as of mid-2026, not verified against vendor documentation); three expert reviews, each reading its assigned chapters in full and running R, Quarto, and Git checks where feasible; and a structural reading of *Advanced R* (2nd edition) as a model of presentation (Part 5, sections 5.5 to 5.9). The expert reviewers were simulated personas instructed to review as a senior pharmaceutical biostatistician, an academic biostatistician, and a senior data scientist would; they are not external human reviewers, and Phase D's catalog listing should wait for human expert review (Part 10).

`DEEP-REVIEW.md` (2026-07-05) covered voice, disciplinary framing, and peer comparison, and `REVIEW-CHECKLIST.md` (2026-09-07) is the per-chapter ownership review. Neither is superseded.

Executive summary

Where the structural work stands. The September plan found the book in good condition on the criteria reviewers rate most often and in poor condition on three that matter most for learning and adoption: exercises without solutions, code that did not execute, and a reader-facing license (CC BY-NC-ND) that disqualified it as an open resource. Most of that has been fixed. The prose is now CC BY 4.0 with all declarations reconciled; the PDF has a table of contents; every rendered figure has alt text; executable chunks rose from 48 to 166; 14 chapters publish worked solutions and the other 20 state why they cannot; a glossary exists; and a `renv.lock` was added. Open items are the index, three of five portraits, the catalog listing and print-on-demand, CI that actually executes code, and a Dockerfile. The `renv.lock` has never been restored: on 2026-10-09 the PDF render failed against the empty project library and succeeded only with renv disabled.

What the reading found. The book has a strong intellectual spine and weak delivery to its stated audience. The spine is the idea that the dangerous errors in applied statistics are *silent*: a join that multiplies rows, a package update that changes a number, a stale intermediate file, a figure that misleads without a false value. That idea recurs in at least fourteen chapters, is never named, and is rarely what a student meets first. Six problems account for most of the readability gap.

1. **Every chapter makes the student wait.** An epigraph, a 'Sources' or 'Why this chapter exists' note, a bypass quiz, and four to seven objectives precede the motivating story, which typically arrives 40 to 90 lines in. Several of those notes address curriculum reviewers rather than students.
2. **The chapter template repeats itself.** 'Orientation', 'The statistician's contribution', and 'Principle in use' restate the same points in most chapters.
3. **The book does not use its own conventions.** `conventions.qmd` promises Verification, Pitfall, and LLM-prompt callouts. None appears in any chapter (verified).
4. **There is no running example.** Successive chapters use penguins, airquality, band_members, a 'read-missions' cohort that never exists as data, and several unrelated synthetic schemas.
5. **The register sits above the audience.** Formal first-person plural, hedging, and undefined acronyms raise the cost of every page. This is the point on which the reading most directly questions an earlier decision (the rgtvoice pass); see Part 11.
6. **Correctness defects cluster where code does not run.** Examples include an underpowered model SAP, a daylight-saving demonstration that prints the opposite of its comment, and a merge-conflict lesson whose 'merged' answer is identical to one side.

The remedy for the readability gap is not new content. It is a smaller, consistent chapter template, one running study carried from the first chapter to a capstone, executed examples, and a short numbered set of habits.

What the expert reviews found. Three expert reviews (pharmaceutical biostatistics, academic biostatistics, industry data science) each concluded that the book should be adopted only after major revisions, and for overlapping reasons. The statistical core is wrong where students cannot check it: the missing-data chapter inverts the condition under which complete-case analysis is unbiased and rests its conclusions on a simulation that reseeds itself into a handful of datasets; the model SAP is underpowered, misdescribes its estimand, and the HF-HOME example contradicts itself across three chapters. Several compliance statements would be harmful if acted on (PHI deposited in dbGaP; date shifting called Safe Harbor; 'FDA requires SAS'). Many infrastructure recipes do not run as written, CI cannot fail, and the LLM guidance contains no data-governance rule. The reviewers also agreed on what is distinctive and worth keeping: the statistician's judgment layer, the review checklists, acquisition, testing philosophy, and communication.

Order of work. Tier 1 defects (Part 6, Phase E1) come first and should be fixed before the book is assigned again. Then the remaining correctness work, the template and running study (the author's decisions, recorded in Part 11, fix their scope), and finally the chapter-by-chapter rewrite, in which the content gaps the reviewers identified (Phase K) are added as sections paired with cuts, so that the book does not grow.

Part 1. Measured state

Baseline is the 2026-09-07 scan; 'Now' is 2026-09-08, after Phases A to D. Rows without a 'Now' value are unchanged.

Property	Baseline	Now	Target from evidence
Total words	111,865	~120,000	n/a
Pages	~248 (estimated, wrong)	**712** (rendered)	Downey: ~140 per semester course
Chapters and appendices	38	39 (glossary added)	n/a
Median chapter, rendered	n/a	**18 pages**	one sitting
Chapters over 25 pages	n/a	4	split or justify
Exercises	173 in 34 files	176 in 34 files	n/a

Property	Baseline	Now	Target from evidence
Chapters publishing solutions	0	**14**, 48 chunks	OpenIntro: nearly all
Chapters labeling open-ended sets	0	**20**	state why no key exists
Executable `{r}` chunks	48, in 11 files	**166** (168 on 2026-10-09, including solution chunks; Phase C's '118' is an interim figure)	code that renders is code that is checked
Top-level ```` ```r ```` display blocks	128	**60**	mostly correct as display
Technical figures (code-generated)	~20	27	1 per code-bearing section
Portraits placed (policy allows 4 to 5)	0	**2** (Gentleman, Wickham; commit `57b9bc5`)	5, per the April policy
`credits.qmd` rows complete	0 of 5, plus cover TODO		all
Other ambient imagery	0		0, per the April policy
Technical sections	374		n/a
Of those, code-bearing	170		each yields a rendered artifact
Of those, prose only	204		exempt from the figure floor
Files whose only figure is a mermaid diagram	29		n/a
Content files with zero figures	5		zero
`fig-alt` coverage	0 of 46 files	**all rendering figures**	all informative figures
Figure labels inside display fences	5, in `11-quarto`, `12-rmd-workflow`, `23-penguins`		render nothing; see Finding 2
Files containing `fig-cap`	34 of 46		n/a
Embedded video, iframe, or media	0		distributed through chapters
Glossary	absent	**appendix F, ~75 terms**	present
Index	absent	absent (see Phase D)	present
HTML table of contents	docked sidebar, search, breadcrumbs		correct as is
PDF table of contents	absent (`toc: false`)	**present, verified**	present
Quiz-to-answer link	present in all 34	unchanged	adjacent
License, reader-facing	CC BY-NC-ND 4.0	**CC BY 4.0**	CC BY
License, code	MIT / CC0 / GPL-3, disagreeing	**MIT printed, GPL-3 package**	code only
Body-text contrast	~16:1; teal accent ~5.0:1		at or above 4.5:1
Heading hierarchy	clean, no level skips		correct as is
Output formats	HTML, PDF (6.5 x 9in trim)		plus print-on-demand
`freeze: auto`	set		correct as is
Bibliography entries	99; 34 from 2024 or later		correct as is
'Check your understanding' sections	1 per chapter (2 in `03`)		1 per major section

Several things in that table are worth stating plainly as strengths, because they are unusual. The bibliography

is current, with 25 entries from 2024, 4 from 2025 and 5 from 2026, and only 7 entries predating 2008, so the staleness signal that drives de-adoption is absent. The HTML navigation is complete and would draw no reviewer complaint: docked sidebar, full-text search, breadcrumbs, page navigation, and anchor sections. Every chapter opens with `Sources`, `Prerequisites` and `Learning objectives`, which is a real answer to modularity, the weakest criterion in the reviewer corpus. (*2026-10-09:* the reading pass qualifies this. The modularity function is real, but the same apparatus delays every chapter's motivation, and two chapters open differently; Finding 11 and the template in 4.1 keep the function in a shorter chapter card.) Contrast and heading hierarchy pass their accessibility checks. And the book already has a per-chapter prerequisites quiz with published answers, which is a retrieval practice structure most textbooks lack entirely.

[**corrected 17:40 PDT**] The first draft credited that quiz structure without auditing where its answers sit. They sit at the last heading of every chapter. In `13-wrangling.qmd` the quiz is at line 25 and `Prerequisites answers` at line 692. The structure is therefore right and its placement is the specific defect the reviewer corpus names, namely solutions that require scrolling to the end of the chapter. The refactoring extends the structure and must also move it. (*Partly retracted 18:05 PDT; see Finding 1. Every quiz already links to its answers, so the defect is minor.*)

Part 2. Findings

Findings 1 to 10 come from the structural pass of 2026-09-07 and carry their implementation status inline. Findings 11 to 23 come from the reading pass of 2026-10-09, Finding 24 from the LLM-section review, Findings 25 to 30 from the expert reviews, and Finding 31 from a visual-density measurement. Where a later finding overlaps an earlier one, the later finding says so.

Finding 1. 173 exercises, zero solutions, and the answers that do exist are as far from their questions as the file allows

Measured. Thirty-four files carry an `## Exercises` section, with 173 numbered items among them, 27 of those files having exactly five. A search for solution or answer headings returns only the per-chapter `Prerequisites answers` sections, which answer the opening quiz and not the exercises.

[**corrected 17:40 PDT, and partly retracted 18:05 PDT**] Those answer sections are the last heading in every chapter that has one, and the 17:40 revision called that placement 'maximally distant' and a direct instance of the reviewer complaint about solutions requiring a scroll to the chapter end.

That was overstated, and executing Phase A disproved it. All 34 chapters already carry an inline cross-reference from the quiz to its answers: 'You can find the answers at the end of the chapter in @sec-quiz-answers-NN.' In HTML that is one click. In PDF it renders as a numbered section reference. The reviewer complaint is about books where the reader must scroll to find the answer; here the reader follows a link. The remaining defect is real but minor: the answer is not adjacent, so the reader leaves the question to reach it.

The book therefore has one substantial defect here, not two. The exercises have no answers at all. The quiz answers are correctly written, correctly linked, and merely not adjacent, which is a refinement rather than a fix. Phase A item 4 was satisfied before it was written.

Why it ranks first. Dunlosky and colleagues rate practice testing as one of only two high-utility study techniques, and the mechanism depends on feedback: a student who cannot check an answer is not doing retrieval practice, only doing homework. OpenIntro ships more than 550 exercises with solutions to nearly all in-text exercises, placed in footnotes on the explicit principle that the check should be immediate. Reviewers in the Open Textbook Library corpus complain when solutions are merely at the end of the chapter; this book does not have them at all. For a self-studying reader, which an open book acquires in quantity, an exercise without an answer is close to worthless.

Complication specific to this book. Roughly half the exercises instruct the student to work on their own project or their own data, which admits no answer key. That is a defensible design for a practicum, but it cannot be the whole exercise set, because it means a reader without an ongoing project cannot self-assess anywhere in the book.

Action. Split the exercise set in two. Retain the open-ended 'apply to your own work' items, and mark them as such so the reader knows no key exists. Add, to every chapter, three to five exercises that operate on a dataset the book ships, and publish worked solutions for those. Place solutions where OpenIntro places them, which is close to the question, either in a collapsible callout immediately following the exercise or in a per-chapter solutions section reached by anchor link rather than by scrolling.

Finding 2. 258 display-only code blocks against 48 executable

Measured. 258 fenced blocks across 27 files marked as plain `r`, `bash`, `sh`, `sql`, `yaml`, `python`, or `sas`, against 48 executable `{r}` chunks confined to 11 files. The first draft reported 226 for the display-only count under a narrower regex; 258 is the figure to work against. The display-only blocks are not chunks Quarto declined to run; they are blocks that were never wired to run. The worst concentrations are `22-sas` (25), `11-quarto` (22), `12-rmd-workflow` (19), and `05b-git-teams` and `13-wrangling` (17 each).

Why it matters. Two consequences follow, and the second is the expensive one. First, none of that code is verified by rendering, so a typo in a display-only block survives every build indefinitely, and `freeze: auto` does not help because there is nothing to freeze. Second, code that does not execute produces no output, which is the direct upstream cause of Finding 3. A chapter on plotting that shows plotting code without plots is the canonical reviewer complaint in this literature.

Corrected triage, 19:10 PDT, and it shrinks the problem by more than half. The 258 figure counts every non-executable fence in the book, and most of them are correctly non-executable. Parsed properly, with nested fences handled:

Fence	Count	Disposition
```` ```r ````	128	the only convertible pool
```` ```bash ````	80	stays: git, Docker, shell, cloud
```` ```yaml ````	15	stays: configuration examples
```` ```sas ````	15	stays: no SAS at render
```` ```qmd ````, `Rmd`, `dockerfile`, `json`	20	stays: displayed source

So 130 of the 258 must never convert. A further 18 ```` ```r ```` blocks sit *inside* ```` ````qmd ```` examples and are displayed source too, which is why a naive `grep -c '^```r$'` returns 146 rather than 128.

Observed conversion rate, from two chapters actually done: 58%, not the ~90% this plan assumed. In `13-wrangling`, 9 of 17 converted; in `14-types`, 9 of 14. The blocks that resist are a consistent set: deliberate antipattern pairs marked `# bad` and `# good` where the bad branch must not run, file I/O against paths the book does not ship, and environment-dependent calls such as `Sys.setlocale` to a French locale. These are not defects. A book teaching what not to do needs code that does not run.

Applying 58% to the remaining 110 gives roughly **64 further conversions**, for about 82 in total rather than the 118 implied earlier. Re-measure rather than trusting that extrapolation; two chapters is a thin base.

Realistic scope. Much of this code genuinely cannot execute at render: shell sessions, Docker builds, `git` interactions, cloud provisioning, and SAS. That is a legitimate reason for a display block, and those should stay display-only and be verified another way. The target is the subset that could execute and simply is not: the wrangling, types, joins, graphics, missing-data, CDISC, and case-study chapters.

Action. Triage all 258 blocks into three categories. Executable and should run, convert to `{r}` chunks. Not executable in principle, mark explicitly and add a verification note saying how the command was checked and when. Executable but expensive, run once and cache with `freeze`. Do the conversion chapter by chapter, verifying the render after each, since converting a block that errors will break the build.

Finding 3. Twenty-nine chapters whose only figure is a schematic, five files with nothing at all, and no separation of technical from ambient imagery

Measured. [**corrected 17:40 PDT**] Eleven files contain executable `{r}` chunks: `02b-deidentification` (2), `12b-acquisition` (7), `13-wrangling` (4), `14-types` (1), `15-joins` (2), `15b-databases` (7), `16-graphics` (6), `17-missing` (2), `19-cdisc` (6), `23-penguins` (8), and `25d-ethics` (3), for roughly 20 computed figures in total. Twenty-nine files contain exactly one figure, a mermaid diagram.

The first draft credited `11-quarto` with three R-generated figures and `12-rmd-workflow` with two. Neither has any. `grep '^```{r'` returns zero for both, and no `11-quarto_files` render directory exists. The first draft also missed `02b-deidentification`, `15b-databases`, and `14-types`, which do have chunks.

A claimed live bug, retracted 19:10 PDT. The 18:30 revision reported that `23-penguins.qmd` line 470 contained broken cross-references to `@fig-scatter` and `@tbl-coef`. It does not. Line 451 opens a ````qmd` fence, so that entire passage is the displayed *source* of an example manuscript, not live document content. The render confirms it: the text appears escaped inside a code block and Quarto emits no crossref warning for it. The claim came from reading a `sed` window that did not extend far enough back to show the enclosing fence.

Two genuine broken cross-references were found by rendering, and are fixed. `25c-federal.qmd` pointed at `@sec-quiz-answers-02` and `25d-ethics.qmd` at `@sec-quiz-answers-02c`; neither anchor exists, and the correct targets were `-25c` and `-25d`. Quarto reported both as `WARN: Unable to resolve crossref`. These were invisible to every grep in this plan and surfaced the moment the book was built, which is the argument for rendering as a verification step rather than a final one.

The mechanism of that error, confirmed 18:05 PDT. Both files do contain `#| label: fig-` lines, five between them, which is what the first draft counted. Every one of them sits inside a ‘`qmd orr` display fence, shown to the reader as an example of how to write a figure chunk. They are illustrations of syntax, not figures. A scan that greps for `fig-` labels without checking whether the enclosing fence is executable will count them, and a chapter that teaches figure-chunk syntax is exactly the chapter where it will do so. Count figures by executable fence, never by label.

Five content files contain no figure of any kind, neither computed nor schematic: `11-quarto`, `25b-communicating`, `C-alternative-wrangling`, `D-peer-survey`, and `E-lab-software`.

`11-quarto` is arguably the sharpest instance in the book, sharper than `25b-communicating`. It is a chapter about rendering, carrying 22 display-only blocks, that renders nothing. `25b-communicating` is the close second: 4,999 words, the second-longest file in the book, fifteen major sections, no figure, no table, and no code block of any kind.

Also measured, and missed by the first draft. The book contains no embedded video, iframe, or media element anywhere. Distributed video is the one addition students in the design studies asked for by name. This is noted for completeness rather than proposed as work; producing video is a far larger commitment than the rest of this plan and should be decided separately.

Why it matters. ‘The textbook contains only words’ is a verbatim reviewer complaint from the Open Textbook Library corpus, offered with the prediction that it costs student engagement. The mermaid diagrams are genuinely useful and should stay; the point is that a schematic of a workflow is not the same as seeing what the code produced.

Relationship to `DEEP-REVIEW.md`. That document identified this exact defect in July 2026 and reported it addressed ‘in the highest-value chapters’. The measurement confirms the fix was real and partial: the eight chapters with computed figures are presumably those chapters. This plan proposes finishing the job.

Action, revised 17:40 PDT. Two kinds of figure, two different targets.

The first draft used ‘figure’ for one thing. It is two, and they are governed by different evidence, so this revision separates them.

A **technical figure** is generated by code in the chapter: a plot, a rendered table, a printed output block, a diagram of a structure the chapter builds. It shows the reader what the code produced. An **ambient image** is related to the topic but not produced by it: a photograph, a portrait of a scientist, a generated metaphor figure. It sets tone.

The evidence treats these oppositely. Technical figures answer the 'contains only words' complaint and are what reviewers mean when they ask for figures. Ambient images fall under the coherence principle and the seductive-details caution already recorded in Part 9, which say that extraneous material placed inside an explanation can reduce learning. Note that the blog post flags its Mayer and seductive-details citations as unverified against primary sources, so that caution is the weakest-evidenced claim in this plan and is treated here as a reason for restraint rather than prohibition.

Accessibility marks the same divide from the other side. WCAG assigns a decorative image a null `alt`, which is an instruction to screen reader users to skip it. An image that genuinely needs a description is carrying information and is a technical figure. So the two categories are not a matter of taste; they differ in whether a subset of readers is told to ignore them.

Target 1, technical figures: one per code-bearing section. Not one per chapter, and not a fixed count per section. The rule is conditional, because a uniform per-section floor would force invented plots into the prose sections, and an invented plot is a seductive detail with a `ggplot` theme.

Measured: of the 374 technical sections in the book (excluding the eight boilerplate headings that appear in every chapter), 170 contain a code fence of any kind and 204 are prose only. The conditional rule therefore targets roughly 170 rendered artifacts against the current ~20, and exempts the 204 prose sections. The per-chapter distribution is uneven and matches the subject matter, from `11-quarto` at 10 and `16-graphics` at 9 down to `25b-communicating`, `D-peer-survey`, and `E-lab-software` at zero.

`25b-communicating` is the case the rule handles badly and needs a judgment call. It has ten technical sections and no code, so the conditional rule exempts it entirely, which is the wrong answer for the chapter with the sharpest visual deficit in the book. Treat it as an exception: it should acquire three or four rendered artifacts, a memo, a figure it discusses, a reviewer-response table, by having code added rather than by being exempted.

Priority order for the technical work is the files where a visual is load-bearing and currently absent: `11-quarto` (a chapter on rendering that renders nothing, 10 code-bearing sections and no output), `25b-communicating` (as an exception, above), `12-rmd-workflow` (7), `20-testing` (8), `24-adni` (5), `C-alternative-wrangling` (4), and `17-missing` beyond its current two. `14-types`, `15-joins`, `15b-databases`, and `12b-acquisition` already have chunks and need more output rather than a first figure. This work is largely a byproduct of Finding 2 and should be scheduled with it.

Target 2, ambient images: finish the policy that already exists. [revised 18:20 PDT, during Phase A execution] The book already has a written image policy at `analysis/report/images/README.md`, dated 2026-04-23, which neither the 14:05 nor the 17:40 draft consulted. It says 'content figures only inside chapters', that 'figures exist to teach', and: 'Do not add decorative chapter-opener images, AI-generated hero images, or part-divider art.' It permits 'portraits of foundational figures: permitted sparingly (four to five across the book)', requiring CC-BY or CC0 sourcing and a `credits.qmd` entry for each, and it names the five: Jenny Bryan in `05-git-solo`, Ben Marwick in `07-rrtools`, JJ Allaire in `11-quarto`, Yihui Xie in `12-rmd-workflow`, Hadley Wickham in `13-wrangling`.

That policy reaches this plan's conclusion on generated imagery by the same reasoning and is stricter on everything else. It governs. The 17:40 draft proposed an ambient image at each of the eight part openers; that proposal has been dropped. It is forbidden by the policy as part-divider art, and it does not survive the evidence on its own terms either: a part opener is a book-level structure serving a reader who moves through the book in order, and the reading-compliance evidence says that reader is rare. What the corpus asks for is modularity at section and chapter level, which this book already supplies through per-chapter `Sources`, `Prerequisites`, and `Learning objectives`. The orienting work a part opener would do is already done once, by `fig-book-map` in `00-intro`.

The ambient target is therefore the policy's own unfinished business: five portraits, none of which exist. `images/portraits/` is empty, all five `credits.qmd` rows read TODO for source and license, no chapter references a portrait, and the cover image's attribution line is also TODO. This is not a budget to design. It is a budget designed in April and never started.

Action. Source the five named portraits under CC-BY or CC0, place each at the first substantive mention of its subject using the compact right-aligned block the policy specifies, complete the five `credits.qmd` rows with source URL and license, and complete the cover attribution. Add no other ambient imagery.

The one thing the policy does not address is cultural relevance, which at 66.6% positive is the second-weakest criterion in the reviewer corpus and the one ambient use with affirmative support in the evidence base, since the guidance asks for diversified names, contexts, and photographs in examples. Five portraits of tool authors do not serve it. Note this as an open question for the human reading pass rather than resolving it here: the answer may lie in the examples and names rather than in more images.

Generated metaphor figures are excluded for three reasons and not only the seductive-details one. They have no provenance to record, and `credits.qmd` requires attribution for material. They are the purest instance of decoration the category admits. And a book containing both `21-ai-coding` and `25d-ethics` will be read closely on precisely this point, so illustrating it with unattributed generated imagery invites the review one would least like to receive.

Acceptance for both targets. Every code-bearing section produces at least one rendered artifact in the built book, verified by rendering rather than by grep, with `25b-communicating` handled as the stated exception; the five portraits named in the April image policy are placed and fully attributed in `credits.qmd`, the cover attribution is complete, and no other ambient imagery has been added.

Finding 4. The reader-facing license is CC BY-NC-ND, and three other files disagree with it

Measured. [**corrected 17:40 PDT**] The first draft reported that `LICENSE.md` and `DESCRIPTION` both declare GPL-3. That is wrong about `LICENSE.md` and incomplete about the rest. Four files declare a license, in two camps:

File	Declares
`LICENSE.md` line 4	CC BY-NC-ND 4.0
`analysis/report/index.qmd`, License section	CC BY-NC-ND 4.0
`DESCRIPTION`	GPL-3
`CITATION.cff`	GPL-3

`index.qmd` is the landing page of the published book, so CC BY-NC-ND is what a prospective adopter actually reads. It also states that code samples are CC0, which is a third position and the only one of the four that is defensible as written.

Why it matters, and why this is worse than the first draft said. The first draft framed the problem as a software license misapplied to prose, which is a category error with ambiguous consequences. The actual problem is an active, published restriction on precisely the reuse that defines the category. ND forbids derivative works, and revision and remix are the affordances that make a work an open educational resource at all, so CC BY-NC-ND does not merely fail to be accepted by the catalogs, it fails to qualify as an OER. NC compounds it by blocking the print-on-demand economics of Finding 9 and by introducing the compatibility conflicts that make combining sources hard. Note that this plan's own Part 9 instructs against adopting an NC or ND variant, written in apparent ignorance that the book already carries one.

The secondary problem is the disagreement itself. Four files, two answers, and no statement anywhere of a prose-versus-code split. An adopter who checks two of them learns that the authors do not know what the license is.

Action. Unchanged in substance from the first draft, but now a correction rather than a clarification. Relicense the prose under CC BY, keep MIT or GPL-3 for the code in `R/` and the scripts, and state the split in all four files plus the colophon so that they agree. Confirm that every contributor consents before relicensing, since this cannot be undone unilaterally later. Anyone who has relied on the NC or ND terms to date should be identified before the change, though in a repository with no catalog listing and no recorded adopters that population is probably empty.

Finding 5. No glossary, no index, no alt text, and no table of contents in the PDF

Measured. No glossary file or heading anywhere in the book. No index. Zero of 46 files contain a `fig-alt` attribute, against 34 that contain `fig-cap`.

[**corrected 17:40 PDT**] The first draft asserted that the book 'does have a navigable sidebar table of contents, so this is two thirds of a known complaint rather than all of it.' That is true of the HTML output and false of the PDF. `_quarto.yml` sets `toc: false` under the `pdf` format, and no `\tableofcontents` appears in any of the LaTeX includes. The PDF is therefore a 712-page document (rendered and counted 2026-09-08) with no table of contents, no index, and no glossary, which is the complete verbatim complaint rather than two thirds of it, and it is the closest match in the entire reviewer corpus to this book's actual condition: a long PDF with 'little in the PDF that allows for quick and easy browsing without intense scrolling'.

The HTML output is the opposite case and needs no work. It has a docked sidebar, full-text search, breadcrumbs, page navigation, and anchor sections. Body-text contrast is roughly 16:1 and the teal accent used for links and code computes to about 5.0:1 against white, both above the 4.5:1 threshold, and the heading hierarchy contains no level skips.

Alt text is a WCAG requirement for informative images and an accessibility criterion reviewers apply; at zero coverage the book fails it outright in both formats.

Action. Four separable tasks. Set `toc: true` for the PDF format and verify the result renders sensibly under `scrbook`, which is a one-line change and the cheapest item in this entire plan. Add a glossary appendix collecting the terms already bolded on first use across the chapters, which makes this largely an extraction job rather than a writing job. Add a back-of-book index for the PDF output. Add `fig-alt` to every figure, which is roughly 50 short descriptions once Findings 2 and 3 have increased the figure count.

Finding 6. Exercise count is uniform, not proportional

Measured. Twenty-seven of the 34 files with an `## Exercises` section carry exactly five, irrespective of chapter weight. `10b-targets` at 1,882 words has five; `12b-acquisition` at 5,249 words has five. The six exceptions are `03-team-science`, `15-joins`, `20-testing` and `25d-ethics` at six, `12-rmd-workflow` at seven, `19-cdisc` at four, and `C-alternative-wrangling` at three.

Why it matters. This is the structural item flagged in `REVIEW-CHECKLIST.md` Phase 1 as 'exercise placement and count proportionate to the chapter's weight'. Uniformity of exactly five is the signature of a template applied evenly rather than a judgment made per chapter, and it means the longest and most demanding chapters are the least practiced.

Action. Rebalance to roughly one exercise per major section, which would put the short chapters near three and the long ones near eight. Fold this into the Finding 1 work rather than doing it separately.

Finding 7. Retrieval density is one check per six pages

Measured. Almost every chapter has exactly one 'Check your understanding' section; `03-team-science` has two.

Why it matters. The existing structure is right and there is simply not much of it. Given the reading-compliance evidence, that most students do not read a chapter straight through, a single mid-chapter check is easy to miss entirely.

Action. Target one short retrieval prompt per major section, with the answer immediately available. This is cheap relative to its evidential support and should be done alongside Finding 1.

Finding 8. Length and modularity

Measured from the rendered PDF, 2026-09-08. 712 pages across 39 chapters and appendices. Median chapter **18 pages**, mean 17.8. The distribution is tighter than the word counts suggested:

Pages	Chapters
26 to 28	`13-wrangling`, `16-graphics`, `17-missing`, `22-sas`
20 to 24	12 chapters
14 to 18	15 chapters
10 to 12	5 chapters
4 to 8	3 appendices

This changes Finding 8 substantially. The earlier version ranked chapters by word count and named six over 4,000 words as split candidates. Rendered, the picture is different. `25b-communicating` was second-longest by words and is 20 pages, mid-pack. `05b-git-teams` was third and is 18. The genuinely long chapters are `13-wrangling`, `16-graphics` and `17-missing` at 28, none of which appeared in the word-count list at all, because code and figures occupy pages that words do not count.

Median 18 pages is roughly a 45-minute read, which is above the one-sitting target but not alarmingly so, and no chapter is an outlier demanding a split on length alone. **The recommendation is now to split nothing on these numbers.** The book's length problem is its total, not its per-chapter distribution, and the total is a consequence of scope plus the executable-output work of Phase C rather than of any chapter sprawling.

The glossary is the longest single unit at 35 pages, which is expected for a reference appendix and is not a reading unit.

Word counts are more reliable than the page estimates, so they are given directly here. Median chapter 2,900 words. Six files exceed 4,000 words: `12b-acquisition` (5,249), `25b-communicating` (4,999), `05b-git-teams` (4,908), `25d-ethics` (4,762), `22-sas` (4,262), and `03-team-science` (4,044). The first draft omitted `03-team-science` from this list.

Why it matters. Downey's target of roughly 140 pages for a semester course is a guideline from one author rather than an empirical result, and a practicum spanning Git through CDISC has a defensible reason to be longer. I would not restructure the book to hit 140. The per-chapter figure matters more, and at a median of about six pages the book is already close to the one-sitting target. Modularity is the weakest criterion across the Open Textbook Library corpus, at 62.5% positive, and this book is better placed than most because every chapter opens with a prerequisites quiz that states what it assumes.

Action, revised 2026-09-08. Split nothing on length. The rendered median is 18 pages and no chapter is an outlier; the four longest are `13-wrangling`, `16-graphics`, `17-missing` and `22-sas` at 26 to 28, none of which the word-count proxy identified. The book's length question is its 712-page total, not its per-chapter distribution.

Superseded. Split only the five long chapters, at natural seams, and leave the total length alone. Make the modularity that already exists explicit by stating in the preface that chapters are assignable individually, since an instructor assigning three chapters of a free book incurs no cost justification.

Finding 9. Adoption mechanics are entirely unaddressed

Measured. The book renders to HTML and PDF and is hosted at `practicum.rgtlab.org`. There is no evidence in the repository of a catalog listing, an ISBN, a print-on-demand edition, or a list of adopting courses.

One thing the first draft missed, in the book's favor. The PDF geometry is already set to a 6.5 x 9 inch trim with asymmetric inner and outer margins, which is a print book's page and not a screen's. The print-on-demand edition is therefore closer than the first draft implied. It is still blocked, though, on both the NC license of Finding 4 and the missing PDF table of contents of Finding 5, because a 712-page paperback with no way to navigate it is not a product worth selling.

Why it matters. Perceived quality is mediated by colleague recommendation, personal knowledge of the author, and peer review rather than by intrinsic merit, and a commercial publisher's marketing resources are, in OpenIntro's phrase, humbling. A book with no distribution strategy is read by the author's own students and nobody else. Separately, more than a quarter of students in one e-textbook pilot believed they would have learned more from paper, and print-on-demand is nearly free to offer.

Action. After the license change, list in the Open Textbook Library and solicit reviews, which is simultaneously a credibility and an awareness act. Produce a print-on-demand paperback from the existing PDF. Publish a page of adopting courses as it accumulates.

Finding 10. The book does not satisfy its own Five Pillars

Added 2026-09-08. Not in the original scan, and found only by looking at the repository's CI while migrating the testing chapter. `10-zzcollab` names five pillars as the minimum for a reproducible project. This project has two and a half of them.

Pillar	In this repository
1. Dockerfile	**absent**
2. `renv.lock`	**absent**
3. `.Rprofile`	present, but never loaded (below)
4. Source code	present
5. Data	present, synthetic and `palmerpenguins`

There is no `renv.lock` anywhere in the repository. The book that devotes a chapter to `renv`, and whose own figure lists the lockfile as pillar two, does not pin its own R packages. The render now depends on roughly thirty packages, and nothing records which versions produce the numbers in the text.

The `.Rprofile` is inert. It sits at the repository root, but Quarto renders from `analysis/report`, and R sources `.Rprofile` from the working directory. Running R from the render directory shows `RETICULATE_PYTHON` unset, so the file has no effect on any build.

Were it loaded, it would be worse rather than better. It hardcodes `/opt/miniconda3/bin/python3` and instructs the reader to 'adjust the path below to match your machine', which is the ambient-machine-state dependency the surrounding chapters exist to argue against. The `reticulate::py_require()` declarations added in Phase C supersede it.

CI renders from a committed cache and therefore never executes the code. `_freeze/` is tracked, 120 files, and holds fully executed markdown. `.github/workflows/render-book.yml` finds no `renv.lock`, so it installs only `knitr` and `rmarkdown`, then renders. That succeeds only because Quarto reads the frozen results rather than running anything. The workflow is a markup gate, not an execution gate.

This qualifies a claim made repeatedly in this plan. Finding 2 argues that converting display blocks to executable chunks means the code is 'checked by rendering'. It is checked when the author renders. It is not checked by CI, and a package update that broke every chunk would deploy green.

In fairness, a book is not a clinical analysis, the freeze cache is the correct mechanism for an expensive build, and committing it is deliberate. The objection is narrower: the project asks readers to adopt practices it does not apply to itself, and a reader who clones this repository to see a worked compendium finds two pillars missing.

Action, in order of value.

1. Add a `renv.lock`. This is the one that unblocks the others, and it is `renv::init()` plus `renv::snapshot()`.
2. Move `.Rprofile` to `analysis/report/`, or delete it. It currently does nothing, and what it would do is wrong now that `py_require()` handles Python.
3. Add a scheduled CI job that renders with `--cache-refresh` against the restored lockfile, so the code is executed somewhere other than the author's laptop. Weekly is enough; it is the only thing that would catch upstream package drift.
4. Consider a Dockerfile. The workflow already builds and renders in one when present, so this is the cheapest pillar to add.

The following findings come from the 2026-10-09 reading pass. Each states the observation, the evidence, and the principle from the literature that it engages. Line references are to `analysis/report/`.

Finding 11. The hook is buried under scaffolding

Most chapters open in this order: epigraph, metadata callout, three-question bypass quiz, learning objectives, then 'Orientation'. The motivating content usually appears 40 to 90 lines into the source. Examples include the following.

- **`01-why`.** 'They are entropy… A step was done by hand on a Thursday afternoon and never written down' (01:69-76) is excellent and arrives after a boilerplate Sources note, a quiz, and five objectives.
- **`22-sas`.** The real hook, 'R-only graduates from this program have reported being excluded at the screening stage' (22:26-29), is inside a metadata callout. The body opens with a market-share sentence.
- **`02b-deidentification`, `20b-ci`, `25c-federal`, `25-synthesis`, and `00-intro`.** All five open with a definition or a recap rather than a problem.

Several opening callouts justify the chapter to a curriculum committee rather than to a reader: 'This chapter brings the Practicum to parity' (15b:29-30), 'places the Practicum ahead of its peers' (20b:17-30), and the peer-survey notes in 10b and 17. 'Sources' callouts cite a private path (`~/prj/sfw/07-zzcollab/zzcollab/`, 10:20-21), a Dropbox path (19:16-25), and internal blog slugs (06:21-23).

The best hook in the book, the three silent failures 'each caught by an assertion, a lockfile, or a dependency graph', sits in the preface (preface:137-143), which many students will not read.

Principle. Advance organizers and motivation should precede detail. Adult learners engage when material is problem-centered and visibly relevant. On screen, 79 percent of readers scan, and front-loaded content determines whether they continue (NN/g).

Finding 12. Template fatigue and internal redundancy

The roughly seventeen-section skeleton is applied identically everywhere: Prerequisites quiz, Learning objectives, Orientation, The statistician's contribution, content, Worked example, Collaborating with an LLM, Principle in use (always three habits), five exercises, Further reading, and quiz answers. Consistency is a strength for navigation. The cost is repetition, of which the following are examples.

- `05-git-solo` argues 'why Git for solo work' three times before the first command (05:54-144).
- 'The statistician's contribution' repeats Orientation in 15, 16, 18, 19, and 20b, and repeats 'Principle in use' in 10 and 19.
- The CNSR inversion is restated five times in `19-cdisc` and again in `22-sas`.
- The preface, `00-intro`, and `01-why` each restate the federal mandate, the ethics-last rationale, and the audience description.

Principle. Coherence (cut what does not serve the objective) and the NN/g finding that concise text improves measured usability by 58 percent. Repetition that is not retrieval practice is reading cost without learning benefit.

Finding 13. Published conventions are not implemented

Verified by `grep`: `callout-warning`, `callout-important`, and `title='Verification'` occur only in `conventions.qmd`. Every callout in the chapters is either a note (Sources, Check your understanding) or a tip (Solutions). Pitfalls are buried in prose. LLM prompts appear as a plain section in most chapters, as a tip callout titled 'Prompts to try' in 19, and as the never-used 'LLM prompt' callout in the conventions page. `21-ai-coding` describes itself as 'the meta-chapter on the "Collaborating with an LLM" callouts', which do not exist.

The preface and introduction also misdescribe the chapter pattern. They say every chapter opens with 'Sources' (02b and 04b do not), and they promise 'adversarial prompts designed to expose their limits' where the prompts are benign drafting requests.

Principle. Signaling. A small number of consistent visual cues helps readers locate what matters. A convention that is described but not used is worse than none, because it teaches students to ignore the conventions page.

Finding 14. No through-line dataset or cast

The datasets in use, in order, are approximately the following.

Chapters	Data
01, 03, 04, 04b	none, or invented per chapter
02b	four fictional patients (Ana Ruiz and others)
05 to 09	a 'readmissions' cohort that is never real data
10	penguins
10b	a 'visits' pipeline
11	`data/readmissions.csv`, which does not exist
12	`y ~ x1 + x2`
12b	Durham and Chapel Hill CRF extracts
13	penguins and a simulated blood-pressure trial
14	penguins, toy vectors, a synthetic CRF
15, 15b	band_members, then two different patient schemas
16	penguins and a trajectory simulation
17	airquality and penguins
18	a hypothetical trial
19, 22	two independent synthetic ADaM builds
23, 24	penguins, ADNI
25b	HF-HOME trial (1,500 or 2,706 patients; see Part 6)

The preface promises penguins and ADNI as running examples (preface:178-180). Neither appears in Parts I to II.

Principle. A running case is consensus practice (R4DS, Data Science in a Box) and serves spacing and interleaving. Each chapter revisits earlier material on familiar data, so retrieval happens naturally. It also lets later chapters build on outputs students made earlier. 'Real, rich, accessible data' is one of Timbers and Cetinkaya-Rundel's ten rules.

Finding 15. Register and jargon

Examples of the formal register appear in most chapters: 'We would urge the reader to read it instead as two statements' (05b:357-358), 'We bring in this chapter the habits of software engineering to bear' (20:79-80), 'it is, we would argue, never the useful question' (06:58-61), and 'The errors are, we would stress, of the silent kind' (10b:69-70). Clever but opaque phrasing also appears: 'The request does not launder the inference' (25b:122-123) and 'irritating the editor is a strictly dominated strategy' (25b:515-516).

Undefined on first use: inferential reproducibility and BLAS (01), HbA1c and EHR (03), l-diversity and t-closeness (02b), VPC, AMI, vCPU, and SLURM (06), `libgsl-dev` (10), estimand, mITT, and ADSL (19), DUA and Youden index (24), and '243B template' (18, 24), which is a course number.

The best passages already use the right voice. Examples include 'A merge conflict is a disagreement about the analysis, not about the text' (05b:100), 'hosting is not archiving' (25c), and 'The query is the recipe; the dated file is the ingredient' (12b:135). The target register exists in the book. It is not yet the default.

Principle. Personalization: conversational, second-person text improves retention (d = 0.30) and transfer (d = 0.54) in meta-analysis (Ginns, Martin and Marsh 2013). This does not conflict with an academic tone. Precision and second person are compatible.

Finding 16. Exercises assume a career the student does not have

About 15 of the 32 exercises in the first seven chapters require the student's own past analyses, collaborators, PI, or projects. Exercise sets in 05 to 12, 20b, 21, 22, 25b, and 25c say 'no answer key is possible'. Several of

those exercises are in fact solvable against a supplied file (04b exercises 1, 2, and 4, for example). The skills most worth modeling (the memo, the reviewer response, the DMSP) have no model answer anywhere.

Relation to Findings 1, 6, and 7. Phase B already added solutions in 14 chapters and retrieval prompts in most, and it labeled the other 20 exercise sets open-ended because their exercises change machine state, need a collaborator, or need software the book does not ship. That judgment holds for the exercises as written. This finding adds two points. First, it concerns the *premise* of the open-ended exercises (a career the student does not yet have), not only their keys. Second, the running study proposed in Part 5 removes the premise for several of those chapters, which is why Phase G revisits them.

Principle. Retrieval practice with feedback is the strongest effect in the literature (Dunlosky et al. 2013, 'high utility'). Rowland (2014) reports a larger effect with feedback than without. A sequence of worked example, then faded example, then independent problem is strongly supported for novices.

Finding 17. Correctness defects cluster in code that does not run

Part 6 lists the defects. The pattern matters more than the individual items. Chapters 10, 10b, 11, and 12 execute no code, and `24-adni`, the book's showcase case study, is display-only. These chapters contain most of the errors found, including object names that are never created (24:389-424), a pipeline that contradicts its own rule (24:360), and Dockerfiles that differ in base image, pinning, and render command across 07, 08, and 09 and do not render a `.qmd` file. Executed chapters carry a second class of error: hand-typed `#>` output that disagrees with the real output (14:520-523).

For this audience, an error in a worked example is not a typo. A first-year student copies it, gets a different answer, and concludes that either they or the book is unreliable. Both conclusions damage approachability.

Relation to Finding 2. Phase C converted 68 blocks and argued that the remaining display-only code is correctly so. This finding does not dispute that triage. It observes that the defects concentrate in exactly that residue, and in hand-typed output inside executed chunks, which Phase C did not audit. The `tools/check-calls.R` check verifies that named functions exist; it cannot catch a wrong argument, a wrong object name, or arithmetic in prose.

Finding 18. Forward dependencies and sequence

The following are inspected examples.

- **`04-workstation` uses the shell before it is taught.** It relies on `cd`, `stow`, and `.zshrc` before `04b-shell` teaches `pwd` and `cd`.
- **`05b-git-teams` depends on six later chapters**, including resolving `renv.lock` conflicts before renv is introduced.
- **`06-cloud` depends on `09-docker`**, three chapters later.
- **`07-rrtools` requires renv, Docker, and CI** in its objectives before chapters 08, 09, and 20b.
- **'Five Pillars' is undefined where first used.** It appears in 08 and is defined in 10.
- **`10b-targets` comes before `11-quarto`** but its report target presupposes Quarto.
- **`12b-acquisition` invokes the 'raw-data rule of @sec-wrangling'**, which is the next chapter.
- **`03-team-science` claims earlier coverage that has not happened.** It says the book 'has spent several chapters lowering that cost', citing tools taught later.
- **`25d-ethics` comes after the chapters it applies to.** Each of its five failures belongs to an earlier chapter (02b, 12b, 16, 25b), so students meet the failures long after forming the habits.

Principle. Novices cannot hold unresolved references in working memory. Each forward reference is a small split-attention cost.

Finding 19. Encyclopedic and perishable detail inside tutorial text

Tutorial chapters carry reference and version-specific material, including the following.

- AWS AMI IDs, prices, and `pip install awscli` (06).
- ODA registration clicks (22).
- Model strings and the ellmer API (21).
- A full `_quarto.yml` dump and website configuration (11).
- Starship prompts, tmux bindings, and Nerd Fonts in a chapter that promises a 'minimum viable workstation' (04).
- A full `rows_*` tour (15).
- The zzcollab bundle and extension mechanics (10:368-424).
- Dated policy statements ('by the end of 2025') throughout 25c.

Principle. Diataxis separates tutorial, how-to, reference, and explanation. Wilson defines a maintainable lesson as one 'cheaper to update than to replace'. Perishable detail inside narrative text is the classic failure of practitioner-written books.

Finding 20. Length per sitting

Finding 8 measured the rendered book at 712 pages and a median of 18 pages per chapter. Five chapters exceed 25 minutes of reading at 200 words per minute: 12b (30), 25b (29), 17 (28), 25d (28), and 05b (26), with 16 and 22 next at about 23. The personalization effect decays in lessons longer than about 35 minutes. NN/g and segmenting research both favor shorter, self-contained units.

Relation to Finding 8. Finding 8 measured pages and concluded that no chapter is an outlier on length alone, and recommended splitting nothing. Word-based reading time and rendered pages rank chapters differently, because code and output occupy pages without adding reading time. The two measures agree that no chapter is extreme. This finding therefore proposes removing reference and perishable material (Finding 19) rather than splitting on length, with one content-based exception; see Part 11, decision 2.

Finding 21. Memorable devices exist but are scattered

The book already contains strong devices. These include the Five Pillars, the three-legged stool, 'snapshot writes, restore reads', 'convert to factor late', '`collect()` late', 'Estimate, then double', 'Stale results are worse than slow ones', and 'What catches them is not care but arithmetic'. Also memorable are the and/but/therefore skeleton, the five ethics failures and their 'where caught' table, and the 86,400-fold upstream-drift story. None is collected, numbered, or referred back to. No chapter has a cheat sheet. The 'Principle in use' triads, three per chapter for about 35 chapters, add up to more than 100 'habits', which is too many to remember.

Finding 22. The capstone does not integrate

`25-synthesis` is a recap of the table of contents plus a reading list. Its two habit lists contradict each other, and one contradicts the testing chapter. Its claim that 'everything else flows from' the compendium contradicts the ethics chapter's thesis that infrastructure 'is orthogonal to whether the analysis misleads'. It never returns to the opening problem of `01-why`, and it sets no integrative task.

Finding 23. The glossary is disconnected

Appendix F has about 75 terms but no chapter links to it (verified by the reader with `grep sec-glossary`). Bold first uses are not connected to entries. Many terms used in late chapters are missing, among them DUA, AUC, data leakage, MCI, DMSP, dbGaP, HIPAA, IRB, and minimum clinically important difference.

Relation to Finding 5. Phase D built the glossary and noted it was cross-referenced to the chapters that develop each term. The reverse direction, from the chapter text to the glossary, is what is missing.

Finding 24. The LLM sections describe the chat-window practice of about 2024

Added 2026-10-09 from a dedicated review of all 33 'Collaborating with an LLM' sections (plus the 'With AI assistance' callout in `19-cdisc` and the tooling sections of `21-ai-coding`), read in full. Statements about the source are inspected. Statements about current tools and practice are from expert knowledge as of mid-2026 and are not verified against vendor documentation.

The sections are the book's most consistent structure: in every chapter, three prompts, each with 'What to watch for' and 'Verification'. They total about 1,400 source lines, roughly six percent of the book. Several are excellent: `25b` (the model's agreeableness as the vehicle for overclaiming, and adversarial review of one's own draft), `25d` ('the one it will get wrong'), `12b` (ask the model to argue the case *against* scraping), and `05b` (the force-push warning, `--force-with-lease`, and 'revert rather than rewrite'). The weaknesses are systematic rather than local.

1. **The workflow is copy-paste into a chat window.** Nearly every prompt begins 'Paste X and ask Y'. By 2026 frontier practice for analysis work is agentic: Claude Code, Codex CLI, Gemini CLI, Cursor, and Positron Assistant read the project, run the code, observe the output, and iterate. Most of the 'Verification' steps in the book (run the tests, re-render, count rows, rebuild the image) are now steps an agent performs itself. The human's role has shifted to specifying acceptance criteria before the work and reviewing evidence and diffs after it. Neither is taught.
2. **There is no data-governance rule anywhere.** Prompts instruct the reader to paste the first twenty lines of a file (12b), five rows (13), 'the tables' (15), a lockfile, a schema, and in `24-adni` material from a cohort governed by a data use agreement. Neither the sections nor chapter 21 says what may be sent to which endpoint: schemas and synthetic rows yes; PHI, DUA-restricted, or proprietary sponsor data only to an institution-approved deployment with appropriate agreements (zero retention, BAA), and often not at all. The two expert reviews (Part 6) reached the same conclusion independently.
3. **Capability claims are generic and will date.** About twenty sections open with the same formula, 'LLMs handle X well; the trap is Y', and many rely on 'the LLM should know this' or 'in our experience'. These are unfalsifiable and describe models of two generations ago. The durable 2026 failure modes are different: not syntax or hallucinated functions, which current models rarely produce in mainstream R, but plausible analytic choices made silently on underspecified requests; agreeableness (sycophancy); confident claims about policy, regulation, and package versions past the knowledge cutoff; fabricated citations; over-engineering; and, in agentic use, weakening or editing tests until they pass.
4. **The prompts are one-liners.** Best practice is to supply context (the data dictionary, the SAP section, the house style), state constraints, define the acceptance check first, and ask the model to list its assumptions and ask clarifying questions before writing code. No section shows a fully specified prompt.
5. **The highest-value techniques are absent.** These are: writing the assertion or test first and asking the model to satisfy it; independent re-derivation in a fresh session or with a second model as a check, which is the AI analogue of double programming and maps directly onto pharma QC practice; a project instruction file (`AGENTS.md` or `CLAUDE.md`) that encodes the book's own habits (native pipe, `relationship =` on joins, `tinytest`, never edit `data/raw/`); giving the model live access to documentation and the R session through tools such as the `btw` package and MCP servers, which removes most hallucinated-API errors at the source; and committing or branching before any agent touches the repository.
6. **There is no LLM provenance habit.** For a book whose thesis is reproducibility, it never asks the reader to record the model, date, and prompt that produced a piece of code or prose, and it does not mention journal disclosure policies (ICMJE and most publishers now require disclosure, and an AI cannot be an author). Model outputs are nondeterministic and model identifiers retire, so an unlogged LLM step is an unreproducible step.
7. **Volume and repetition.** Three prompts times 33 chapters is about 100 prompts, many of them trivial ('ask for a config; test it; iterate') with verification steps that are not actionable ('have a colleague read it', 'iterate'). One or two strong prompts per chapter would teach more.
8. **Format is inconsistent.** Plain sections in most chapters, a tip callout in 19, and an 'LLM prompt' callout defined in `conventions.qmd` and never used (Finding 13).

Specific errors and weak advice in individual sections are listed in Part 6 and in 4.6.

The following findings come from three independent expert reviews commissioned on 2026-10-09, each reading its assigned chapters in full: a pharmaceutical-industry biostatistician (10 chapters), an academic biostatistician who directs an MS program (14 chapters), and a senior industry data scientist and engineer (19 chapters). Each reviewer ran R, Quarto, and Git checks where feasible, and labeled claims as verified, inspected, or expert knowledge. All three reached the same verdict: **adopt only with major revisions**. Their individual defects are consolidated in Part 6; the findings below are the patterns.

Finding 25. The statistical core is wrong in the places students cannot check

The academic and pharma reviewers independently identified the same cluster, and both verified it by computation.

- **Missing-data theory is inverted (17, repeated in 23).** The book teaches that complete-case analysis is biased under MAR. For regression coefficients, complete-case analysis is unbiased whenever missingness is independent of the outcome given the covariates, whatever the mechanism label (White and Carlin 2010; Hughes et al. 2019). The academic reviewer's 300-replicate run of the chapter's own design gave complete-case bias of -0.003 against multiple-imputation bias of +0.064 when x is missing on x itself, and -0.185 against -0.024 when missingness depends on y (verified). The table also says MAR needs no sensitivity analysis, contradicting the chapter's own principle and regulatory expectation.
- **The simulation behind the chapter's conclusions is broken.** `mice(seed = 1)` inside the replicate function resets the global RNG, so the 100 'replicates' are a handful of datasets (verified by both reviewers: two alternating values in one run, five distinct datasets in another). The interpretation at 17:917-945 rests on it.
- **The model SAP is underpowered and its estimand is misdescribed (18).** It needs 1,094 per arm for 80 percent power, not 750 (verified three times), and attrition is not applied (71, not 64). A covariate-adjusted model with a random site intercept estimates a conditional, not marginal, odds ratio. The analysis regresses on treatment received, not randomized assignment. There are no estimand attributes or intercurrent events, though ICH E9(R1) is cited and death before readmission is the obvious one. The multiplicity scheme (primary at 0.05, then Bonferroni over secondaries) allows familywise error near 0.10; the conditional HC1 rule is a pretest procedure (Long and Ervin 2000 support unconditional HC3).
- **The HF-HOME running example contradicts itself across 03, 18, and 25b.** Trial size (1,500 or 2,706), power target (4.5 or 5 points), estimand (odds ratio or risk difference), and missingness (6.3 percent missing, or 'every outcome ascertained') all disagree. The sensitivity code adds imputed events without enlarging the denominator. The observational intake version in 03 builds in immortal-time bias (exposure 'within 7 days' against a 30-day outcome), uses a causal verb, and records 'Red flags noted: None'.
- **Smaller verified errors:** the daylight-saving example (14), factor joins said to match codes (they match labels, 14), a p-value inconsistent with its interval (03), a confidence interval for a mean presented as an individual prediction (03), and others in Part 6.

This finding strengthens Finding 17 and changes its priority. The running study proposed in Part 5 must be built from *one* canonical SAP with every number computed, or it will propagate these inconsistencies into every chapter.

Finding 26. Compliance and regulatory statements a student could act on are wrong

From the pharma reviewer, confirmed against the source where marked.

- **PHI deposited in dbGaP and TCIA** in the model DMSP (25c:414-419; inspected). Both accept de-identified data only, and dbGaP is not a home for general clinical CRFs (BioLINCC, Vivli, ICPSR, and the NIA and NIDA archives are).

- **Date shifting and MRN hashing presented as Safe Harbor** (02b). Safe Harbor permits the year only; date shifting is an Expert Determination technique; a re-identification code may not be derived from an identifier (45 CFR 164.514(c)). HIPAA and the Common Rule notion of 'coded' data are conflated, and the limited data set with a DUA, the most common research pathway, is absent.
- **'A regulatory submission to FDA requires SAS-formatted deliverables'** (22:98-103; inspected). FDA requires the open XPT transport format, not SAS software, and R-based submissions have been accepted (R Consortium pilots). This misinforms career choices.
- **Industry roles and analysis sets are misdescribed** (19): in industry, statistical programmers build ADaM; mITT is described with the usual safety-set definition; analysis sets are conflated with the estimand's population; `1L - CNSR` silently fails for CNSR values other than 0 and 1 (verified).
- **The DMSP is not 'graded' in peer review** (25c), and the NIH 2024 Public Access Policy is missing.

Finding 27. Many infrastructure recipes do not work as written

From the data-science reviewer, most verified by running them.

- **The PDF prints ``` literally** wherever inline code was escaped (07:330, 11:156, 11:588-595, 12:615; verified with `pdftotext` on the 2026-10-09 PDF).
- **`quarto convert paper.Rmd -o paper.qmd` writes a Jupyter notebook** (12; verified, Quarto 1.10.18).
- **`tinytest::test_all()` exits 0 on failure**, so both CI workflows in 20 and 20b stay green when tests fail (verified); 20b never installs Quarto.
- **`renv.lock merge=ours` does nothing without a driver, and with one silently discards the collaborator's lockfile; `--theirs` inverts under rebase** (05b; verified in a throwaway repository).
- **A popped stash cannot be recovered from the reflog** (05; verified); reflog expiry for unreachable entries is 30 days.
- **The zzcollab chapter does not match the zzcollab repository**: no `modeling` or `shiny` profile, a different bundle schema, no `make render`, and `make check-renv` edits `DESCRIPTION` rather than auditing (inspected against `templates/bundles.yaml` and the Makefile).
- **Docker** (`version =` argument, `rmarkdown::render()` on `.qmd`, mismatched mounts, no `.dockerignore`), **renv** (snapshot type, sandbox, static dependency discovery), and **freeze** (per document, not default, ignored on single-file render) defects, as already recorded in Finding 17, were independently confirmed.

Finding 28. Security and currency gaps for an industry hire

- **Security advice:** a public dotfiles repository as 'the modern standard' (04); IAM users with long-lived keys rather than SSO and instance roles (06); RStudio exposed on all interfaces with a literal password (09); a committed database password (09); `.gitignore` that omits `.Renviron`; no secrets management, no parameterized SQL (15b), and no LLM data governance anywhere (Finding 24).
- **Absent in late 2026** (verified by search): devcontainers and Codespaces, pre-commit, keyring or secrets managers, DuckDB and Parquet as defaults, Posit Package Manager dated snapshots, uv, supply-chain basics (pinning Actions by SHA, `permissions:`, OIDC), Positron as a primary IDE, SLURM batch scripts and job arrays.
- **Industry content absent** (pharma reviewer): MMRM and reference-based multiple imputation; group-sequential designs and DMCs; multiplicity procedures; safety analyses (TEAE, MedDRA); Define-XML, Pinnacle 21; TLF shells and double programming; GxP, 21 CFR Part 11, validation of R (R Validation Hub, `riskmetric`); the R Consortium submission pilots; `rtables`/`tern`, `Tplyr`, `gtsummary`.
- **Academic content absent:** the ASA Ethical Guidelines (2022), Belmont and the revised Common Rule, research misconduct and questionable research practices, causal-inference foundations (DAGs, target-trial emulation; Hernán 2018), Gamble et al. (2017) for SAPs, TRIPOD+AI for the ADNI prediction model, CONSORT and SPIRIT 2025, `pointblank` or `validate` for data validation.

Finding 29. In-house tools are positioned as requirements

All three reviewers raised this independently. Mention counts: zzcollab 68, zztable1 20, zzlongplot 18, zzpower 15 (verified). None is on CRAN. Learning objectives require `zzpower` (18:49); `zzc analysis` is called 'non-negotiable' (23:585-587); `zzrenvcheck` is placed ahead of `renv::status()` (08); the test framework follows from what zzcollab ships (20:82-84); `theme = 'fda'` implies an FDA figure standard that does not exist (24). Appendix E's disclosure is commendable but does not offset the positioning. Industry uses `pwr`, `rpact`, `gsDesign`, `gtsummary`, `rtables`, `ggsurvfit`, and renv with Docker. This strengthens house rule 10 and decision 5 in Part 11 and extends decision 5 to all `zz*` packages.

Finding 30. Scholarship and positioning

- **Reproducibility scholarship is thin** (01): Duke and Potti (Baggerly and Coombes 2009), Reinhart-Rogoff (Herndon, Ash and Pollin 2014), and NASEM (2019) are absent (verified by search); 'inferential reproducibility' is narrowed; data-sharing mandates are conflated with computational reproducibility.
- **Comparator texts are not cited** (verified by search): Alexander, *Telling Stories with Data* (the closest competitor), The Turing Way, OpenIntro, Data Science in a Box.
- **Chapters 13 to 16 largely restate R4DS 2e** with the same dataset. The academic reviewer recommends pointing to R4DS for verb tours and keeping only the clinical-judgment layer, which is the book's distinctive contribution.
- **Prerequisites and pacing are inconsistent:** the preface asks for undergraduate statistics while `00-intro` assumes GLMs and survival analysis, which first-year students are learning concurrently; 35 chapters at two per week is 17.5 weeks, not a quarter.

What the reviewers would keep. The 'statistician's contribution' framing, the analysis-diff review checklist (05b), the semantic- conflict callout, `git bisect run` with an assertion, 12b as a whole, the '500 rows proves nothing' callout, the structural-versus- correctness testing distinction (20), the tag-versus-digest callout (09), the 'you cannot test MAR' callout, precise versus inconclusive nulls and the response-to-reviewers material (25b), the overlapping-interval demonstration and Obermeyer case (25d), GitHub-versus-Zenodo (25c), SAP tag provenance, and the honest disclosures in 21 and appendix E. Several of these ('statistician's contribution') are sections the readability findings proposed to fold away; Part 4 keeps their content and moves it, and the plan should take care not to lose it.

Finding 31. Few figures and tables, and long unbroken prose where there is no code

Added 2026-10-09. Measured from source with `scratchpad/vis/vis.awk`; counts are verified, the thresholds are a working heuristic.

The finding. The book breaks up its text often enough where there is code, but it rarely uses figures or tables, and where code is absent (every chapter's opening and the conceptual chapters) readers meet long stretches of unbroken prose. Specifically:

1. About 21 computed figures with figure labels, concentrated in six chapters (13, 16, 17, 19, 23, 25d); most other chapters have one mermaid diagram as their only image.
2. About 17 tables in the whole book, although its content is full of comparisons, defaults, and checklists.
3. In 15 chapters the longest unbroken prose run begins within the first 20 source lines: the opening scaffolding runs 700 to 1,300 words (four to six minutes) before the first code block or figure.
4. The conceptual chapters are the most text-heavy.

Chapter	Prose words per visual element	Longest run (words)	Run starts at
25b-communicating	902	2,657	467 (null results, reviewers)
03-team-science	989	1,531	433 (portfolio, communication)
18-analysis-plan	326	1,325	21 (opening)
05b-git-teams	218	1,236	659 (LLM section)
01-why	735	1,210	154 (anatomy, failure modes)
25c-federal	744	1,208	19 (opening)
17-missing	178	1,152	627 (reporting)
12b-acquisition	269	1,105	19 (opening)
25d-ethics	369	1,078	19 (opening)
10-zzcollab	291	1,070	394 (bundles)
00-intro	1,556	918	158
25-synthesis	693	889	146
Data chapters (13, 14, 15, 16, 20, 22, 23)	83 to 138	506 to 857	mostly the opening

Method and limits. Visual elements are computed figures (labeled figures in executed chunks), mermaid diagrams, images, pipe tables, executed and display code blocks, and callouts. Prose words exclude code, tables, headings, and fenced-div markers. Counting callouts and display-only code as breaks flatters the infrastructure chapters; unlabeled solution plots are not counted as figures (the September scan found 27 technical figures); output printed by executed chunks is not measured. There is no controlled evidence for a specific visual-density threshold. The rule proposed below (house rule 15) is an editing heuristic consistent with the NN/g scanning findings and the segmenting principle, not an evidence-based target.

Relation to Findings 3 and 11. Finding 3 counted figures; Finding 11 described the scaffolded openings. This finding measures the spacing of visual breaks, which is how readers experience both, and adds two points neither covered: the near-absence of tables, and the specific text-heavy stretches.

The fix is a list, not placeholders. Empty placeholder tables or figures inserted into the chapters were considered and rejected: the book renders and deploys from `main`, so placeholders would eventually ship, and inserting slots at a fixed interval produces graphics for their own sake, which Part 9 forbids. Instead, the candidates below are each tied to source material that already exists in the chapter. Most are conversions of prose that is already structured as a list or comparison; none requires new analysis except where noted.

Key. **Type:** T, table converted from existing prose; F, figure from existing or small new executable code; B, before-and-after pair; D, diagram; C, cut instead of illustrate. **Status:** Ready, can be done now from existing content; After E1, depends on a Tier 1 correction (Part 6); After F, depends on the template or habit list (Phase F); Verify, contains dated or expert-knowledge facts to check against primary sources first.

Chapter:lines	Type	Candidate	Source content	Status
00:37-108	T	'The book in one table': part, the question it answers, key tools; placed beside `fig-book-map`	Eight paragraphs describing Parts I to VIII	Ready
00:218-232	T	Four-step reading method: step, what you do, why it works	Numbered list	Ready

Chapter:lines	Type	Candidate	Source content	Status
00:248-274	T	Chapter anatomy: element, purpose, where to find it	Twelve-item list; rewrite after the template changes	After F
01:96-105	T	Kinds of reproducibility: what is held fixed, what varies, example	Three-item list (correct the 'inferential' definition per Finding 30)	Ready
01:176-178	T	What breaks without each leg of the stool	Three-line list	Ready
01:186-214	T	Five failure modes: how each shows up, which tool catches it, which chapter teaches the tool	Five bold-led paragraphs; also the natural place to name the motto	Ready
01:225-236	T	Cost-benefit scenarios: event, cost without a compendium, cost with one	Three numbered scenarios	Ready
01 (new)	F	Output of the 'break it, then fix it' script run twice	Proposed in Part 7	After G
03:122-188	T	Collaboration spectrum: mode, statistician's role, authorship expectation	Section prose	Ready
03:195-222	T	Authorship scenarios against ICMJE criteria; separate CRediT roles from ICMJE eligibility	Criteria list and scenario list	Ready
03:315-327	T	Scope-of-work template: element, what to write, example	Six bold-led items	Ready
03:335-460	T	A sample project portfolio: project, stage, next deliverable, hours	Portfolio section; breaks the 1,531-word run	Ready
03:461-497	C	Move results communication to 25b	Duplicates 25b	Ready
03:95-102, 499-524	T	Disagreement tiers: preference, best practice, validity, and the response to each	Two overlapping sections, merged	Ready

Chapter:lines	Type	Candidate	Source content	Status
03:534-566	T	The intake document rendered as a filled form	Worked example; remove the immortal-time bias first	After E1
18:57-133	C	Cut the duplicated Orientation and contribution prose	Opening run of 1,325 words	Ready
18:135-153	F	Simulated false-positive rate as analytic choices multiply	‘Researcher degrees of freedom’ argument; a small new simulation	Ready
18:169-204	T	SAP sections: contents, ICH E9 or Gamble et al. (2017) counterpart	Thirteen-item ‘243B’ list	After E1
18:302-315	T	Pre-specified versus exploratory, side by side	Two parallel lists	Ready
18:520-530	F	Power curve for the HF-HOME design marking 750 and 1,094 per arm	Worked SAP; turns the defect into a lesson	After E1
25:37-85	C/D	Replace the eight-part recap with `fig-book-map`, every part checked off	Recap list	Ready
25:149-169	T	The numbered habits: habit, chapter, failure it prevents	Seven-item habit list (reconcile with the map)	After F
25:217-251	T	Reading list by year: title, why	Twelve-item list	Ready
25b:128-174	T	And/but/therefore skeleton filled in for HF-HOME	Narrative-arc section	After E1
25b:217-227	B	Annotated memo: the five parts labeled on a real memo	Five-item memo structure	After E1
25b:235-315	F	Estimate and interval for the primary and each sensitivity analysis	Existing memo chunks, once the denominator is fixed	After E1
25b:428-496	F	Four interval patterns against zero and the MCID: precise null, inconclusive, benefit, harm	Null-results section; synthetic numbers; breaks the 2,657-word run	Ready

Chapter:lines	Type	Candidate	Source content	Status
25b:497-611	T	Three reviewer comments: concede or defend, response, change made	Three worked responses	After E1
25b:612-629	T	Peer-review checklist	Six-question list	Ready
25b (new)	B	Chronological versus answer-first memo, side by side	Described, never shown (Part 7)	After E1
25c:124-147, 197-205	T	Policy 'as of' table: agency, requirement, effective date, source, date verified	Nelson memo and agency lists	Verify
25c:157-171	T	DMSP elements: what to write, HF-HOME example	Six-element list	After E1
25c:180-187	T	Data type, PHI status, appropriate repository, access mode	Data-type list; corrected repositories	After E1
25c:253-294	T	GitHub versus Zenodo versus domain repository: DOI, permanence, access control	Two repository lists	Ready
25d:563-569	T	Move the existing 'where each failure is caught' table to the opening as an advance organizer	Existing table	Ready
25d:148-157, 495-507	T	Aggregation and representativeness checklists	Two three-item lists	Ready
05b:509-550	T	Analysis-diff review checklist as a table	Existing checklist	Ready
05b:339-371	B	Conflict before and after resolution, with row counts	Conflict example, once rebuilt	After E1
05b:665-718	C	LLM section reduced to one callout (4.6)	Source of the 1,236-word run	After F
12:543-567	T	Rendering failures: error message, cause, fix	'Common rendering failures'	Ready
12:166-178	T	R Markdown versus Quarto differences	Four-item list (remove the `quarto convert` claim)	After E1
12b:72-99	T	Source type (file, interface, scrape): risk, how to freeze it	Opening taxonomy; breaks the opening run	Ready

Chapter:lines	Type	Candidate	Source content	Status
12b:277-303	T	Import checklist: pathology, symptom, detection, fix	Five pathologies	Ready
16:115-135	B	Bar of means versus points with summaries, same data	Bar-chart paragraph	Ready
16 (new)	T	Figure checklist: claim, channel, distribution, uncertainty, theme, export	'Statistician's contribution' wall (96-160)	Ready
17:142-172	D	The three mechanisms as small graphs (what missingness depends on)	Mechanism table, corrected	After E1
17:627-643	T	Reporting standards: guideline, item, what to report	Reporting section (item numbers per Part 6)	Verify
17:886-945	F	Bias and coverage of complete-case and MI from the rerun simulation	Solution 2	After E1
20b (new)	B	A red CI log excerpt, annotated: where the first error is	Described in 245-249	After E1
21:202-257	T	LLM failure catalog: failure, example, check that catches it	Nine bold-led modes	Ready
21:337-351	D	The four Cs as a loop	Verify-first workflow	After F
22:549-578	T	Defaults that differ: reference level, coding, covariance, censoring, date epoch	Reference-coding callout and scattered notes	After E1
22:582-626	B	Annotated SAS log: ERROR, WARNING, the NOTE that matters	Log-reading section	Ready
06:204-249, 407-408	T	Compute options and costs 'as of' a date	Service catalog and prices	Verify
All chapters	D	Recurring pipeline diagram with the chapter's layer highlighted (4.5)	New	After F

The 'Ready' items, about 30 of the 54, can be done immediately and independently of every other phase; a handful in the heaviest chapters (01:186-214, 03:499-524, 12b:72-99, 25b:428-496, 25d:563-569) would remove most of the worst runs on their own. The rest are scheduled with the phase they depend on (Part 8).

Part 3. House rules for the readability refactor

These rules translate the literature into decisions an editor can apply mechanically. The evidence strength for each is given in brackets.

1. **Motivation before structure** [moderate]. The first thing under the chapter title is either a three-to-six-sentence problem (a person, a failure, a consequence) or a direct statement of what the chapter will let the reader do (4.1). Metadata moves to the end or to an instructor guide.
2. **Direct, precise, unhedged** [moderate]. Remove 'we would argue/urge/stress/note' and 'it appears' in every case. The author's voice is otherwise kept; instructions, exercises, and callouts address the reader as 'you' (Part 11, decision 1).
3. **Define on first use, link to the glossary** [consensus]. Every bold term links to its glossary anchor. Every acronym is expanded at its first use in each chapter, not only in the book.
4. **One idea per paragraph, conclusion first** [strong for usability]. Any paragraph over about 10 source lines is split or converted to a list or table.
5. **Execute every example that claims an output** [book's own principle]. No hand-typed `#>` lines in executed chunks. Display-only code is allowed only where execution is impossible (cloud, SAS, interactive Git) or would mutate the book's own build (Phase C), and it is then labeled as such; where possible, a self-contained toy project shows real output instead (Phase E2).
6. **Retrieval after every major section, with feedback** [strong]. Use one or two short questions in a collapsed callout directly after the section, not at the end of the chapter. Multiple-choice distractors target named misconceptions.
7. **Exercise ladder in every chapter** [strong for novices]. One faded example and two or three closed exercises with adjacent keys, placed at the ends of the sections they exercise (4.1); then, at the chapter end, one cumulative exercise that reuses an earlier chapter and at most one open 'on your own work' exercise.
8. **At most two callouts per screen, from a fixed set of four** [consensus]. The set is listed in Part 4. No callout may carry prose that belongs in the body.
9. **Perishable facts live in dated tables** [consensus]. Prices, IDs, versions, policy deadlines, and API names go in one 'As of YYYY-MM-DD' box per chapter or in an appendix.
10. **Tool-neutral first, in-house tools second** [consensus]. Each concept is taught with the general tool. `zz*` packages appear as 'one way to automate this', always beside the portable alternative. This makes the claim in appendix E true. It is the minimum under either answer to Part 11, decision 5.
11. **Name the silent-failure rule once and point back to it** [consensus; serves spacing]. See 4.4.
12. **Pilot before publishing** [consensus]. Every rewritten chapter is read by at least two students from the target audience before it is considered done.
13. **One source for every running-example number** [book's own principle; Finding 25]. Every figure quoted about HF-HOME or the running study is computed from one canonical SAP and dataset, never retyped in a second chapter.
14. **Statistical and compliance claims get a second reader** [Findings 25, 26]. Any change to 02b, 17, 18, 19, 22, 24, or 25c is read by a biostatistician or a compliance colleague other than the author before it ships.
15. **Visual spacing** [heuristic; Finding 31]. Aim for at least one figure or table per about 400 prose words and no unbroken prose run over about 600 words. Meet it by converting existing structured prose, by showing output that already exists, or by cutting; never by adding a graphic for its own sake.

Part 4. Redesigned chapter template

4.1 Template

The proposal reduces the repeated scaffolding from about nine sections to five fixed elements, and moves the rest into the body or out of the student-facing text.

Order	Element	Replaces	Notes
1	**Opening** (no heading, 3 to 6 sentences): either a story (a person, a silent failure, a consequence) or a direct payoff statement ('Understanding X will let you: …' with three bullets, the *Advanced R* opener)	Epigraph position, Sources, 'Why this chapter exists'	Keep the epigraph after the opening if wanted
2	**Chapter card**: time to read, 'You'll need' (prerequisite chapters and the `library()` calls), 'You'll be able to' (3 to 4 objectives with measurable verbs), and an **Outline** (one linked bullet per major section)	Learning objectives (4 to 7), Prerequisites prose	Bypass quiz stays, collapsed, with 'if these answers come quickly, skip to the next chapter'
3	**Body**, organized as a few numbered `##` sections with `###` subsections; each major section ends with a 'Check' callout and its own **Exercises** subsection	Orientation, Statistician's contribution, end-of-chapter CYU, end-of-chapter exercises	Statistician's contribution is folded into the sections where each judgment arises
4	**Rules of thumb** (3 or fewer, each numbered against the book-wide list)	Principle in use	See 4.3
5	**Cumulative practice** (one mixed exercise reusing an earlier chapter, at most one open 'own work' exercise), a one-page **cheat sheet**, and Further reading cut to 3 items	Exercises, quiz answers at the end	Cheat sheets also collected in an appendix

Exercises live at the section, not the chapter end. Advanced R closes every major section with its own numbered exercise block (3 to 9 items; 33 across six sections of its functions chapter), some with a '(Hint: …)' scaffold. Placing practice immediately after the material it exercises serves segmenting and spacing better than a single end-of-chapter set, and hints are the cheapest form of fading. Short keys stay adjacent in collapsed blocks; long model answers go to the solutions companion (5.6).

Asides go to margin footnotes, not callouts. Advanced R uses no callout boxes at all; history, version notes, terminology, and side remarks are numbered footnotes. Set `reference-location: margin` in `_quarto.yml` (margin

notes in HTML, ordinary footnotes in PDF) and move version-specific remarks, provenance notes, and tangents there. Together with the four-callout limit in 4.2, this is the main mechanism for clearing the reading path (Findings 11 and 19).

Numbered, nested sections. The chapters are currently flat: in `13-wrangling` every heading, including each 'Check your understanding', is a `##`. Group subsections under fewer `##` headings, so the sidebar is scannable and cross-references can point to a subsection ('as in Section 13.3.2') as *Advanced R* does throughout.

LLM material becomes one small 'LLM prompt' callout placed beside the task it supports, rather than a section. Material that only explains the book's design (peer survey, curriculum parity, Sources) moves to a new *Instructor Guide* appendix or to `D-peer-survey`.

4.2 Callout set

Use exactly four types, and update `conventions.qmd` to match.

Callout	Quarto type	Purpose	Frequency
Check	note, collapsed	Retrieval question with explained answer	After each major section
Verification	tip	A concrete command the reader runs to confirm success	After each worked example
Pitfall	warning	A silent failure students have actually hit	One to three per chapter
LLM prompt	important	A prompt plus the check that verifies its output	Zero to two per chapter

Solutions use collapsed blocks inside the exercise, not a callout type.

4.3 A short, numbered list of habits

Replace the more than 100 'Principle in use' habits with a book-wide list of about twelve numbered habits. Each chapter's 'Rules of thumb' cites the numbers it exercises ('Habit 4: count before and after every join'). The list is printed in the introduction, in the synthesis chapter, and on a tear-out card. The draft below is assembled from rules the book already states.

1. Raw data are evidence; derived data are a build product.
2. Every number in the prose is computed.
3. Snapshot writes, restore reads; pin the environment.
4. Count before and after every join; declare the relationship.
5. Convert late, validate early (types, factors, dates).
6. A bug that produces `NA` is cheap; a bug that produces a plausible number is expensive. Write the assertion.
7. Test the statistics, not the shape.
8. Stale results are worse than slow ones; let the pipeline decide what to rebuild.
9. Pre-specify, then tag the commit before you see the outcome.
10. Hosting is not archiving.
11. Lead with the answer; match the verb to the design.
12. Send the model structure, not data; verify what it wrote. Your confidence measures its readability, not its correctness.

4.4 The book's motto

Name the through-idea once, in chapter 1, and refer back to it by name: **'Make silent failures loud.'** Each chapter's opening story is an instance of a silent failure. Each Pitfall callout names the failure. Each Verification callout is the mechanism that makes it loud. The synthesis chapter closes the loop. The phrase is close to the book's existing language (14:563-565, 10b:69, 11:315, 12:249, 12b:117), so it is a naming exercise, not new content.

4.5 A recurring visual grammar

Advanced R's most transferable device is a single diagram language used across chapters: in its 'Names and values' chapter, 21 diagrams share one grammar (names as rounded boxes, arrows to values, explained conventions for color and dotted lines), which the next chapter extends, and each major section opens with a small summary tree showing where the topic sits. The *Practicum* has about one mermaid diagram per chapter, each in its own style.

Adopt one recurring diagram: the project pipeline from raw data to report, layered with the Five Pillars (raw data, code, environment, document, archive). It appears at the top of each chapter with the layer the chapter addresses highlighted, and grows as the book proceeds. This is dual coding of the book's structure and a 'you are here' marker for readers who enter mid-book. Each instance carries `fig-alt` text, which *Advanced R* omits.

4.6 The LLM callouts (responds to Finding 24)

One protocol, taught once, applied everywhere. Move the durable core of `21-ai-coding` (the verify-first loop, failure modes, data governance, agentic practice) into Part II, so that the protocol is learned before the first chapter that uses it (Part 11, decision 11). Give it a memorable name. A proposal, the four Cs:

1. **Context.** Give the model what a colleague would need: the data dictionary or schema, the relevant SAP section, the house style. Send structure, not data.
2. **Constraints.** State packages, idioms, and prohibitions ('use `relationship =` on every join; do not modify `data/raw/`; no new dependencies').
3. **Criteria.** Write the acceptance check *before* asking: the assertion, the test, the expected row count, the number the rendered document must reproduce. Ask the model to list its assumptions and to ask questions before writing code.
4. **Check.** Run the criteria; read the diff; for anything that will be reported, re-derive it independently (a fresh session, a second model, or by hand). Log the model, date, and prompt.

Rules stated once in 21 and referenced by every callout.

- *Data governance.* Schemas, summaries, and synthetic rows may go to any approved tool. PHI, DUA-restricted data (ADNI), and proprietary sponsor code go only to an institution-approved deployment under the governing agreement, and in many cases nowhere. Check the DUA.
- *Agent safety.* Commit or branch before an agent edits the repository; run agents with the least permission that works; never give an agent production credentials or cloud keys; treat text the agent reads from downloaded files, web pages, and issues as untrusted (prompt injection); verify that any package an agent proposes actually exists on CRAN or PyPI before installing it (hallucinated package names are registered by attackers).
- *Tests are off-limits.* An agent asked to make tests pass may weaken or delete them. Review every change to `inst/tinytest/` separately, or forbid it in the instruction file.
- *Project instruction file.* Ship an `AGENTS.md` (or `CLAUDE.md`) in the running-study repository that encodes the numbered habits. It is the book's habits made machine-readable, and it is the single highest-leverage step a student can take.
- *Provenance and disclosure.* Record model, date, and prompt for any LLM-produced code or text that reaches a deliverable; follow the journal's AI-disclosure policy; an AI is never an author.

- *Live documentation.* Prefer tools that give the model the installed package documentation and the R session (the `btw` package, MCP servers) over relying on its training data.

Per-chapter format. Replace each three-prompt section with one or two 'LLM prompt' callouts placed beside the task they support. Each contains a fully specified prompt (context, constraints, criteria), the chapter-specific failure mode, and a check that is executable, ideally the chapter's own Verification callout. Delete the formula opener ('LLMs handle X well; the trap is Y') and every 'the LLM should know this'. At least three chapters (13, 15, 20) show an agentic version: the agent runs the code and the student reviews evidence.

Chapter actions for the LLM callouts. 'Keep' means retain as an exemplar of the new format after conversion.

Chapter	Keep	Fix or replace	Add
01	Prompt 1 (audit a script)	Let an agent run the script in a clean container rather than guess	Provenance habit introduced here
02b	Prompt 1	Prompt 2 inherits the Safe Harbor mislabeling (Part 6)	Data-governance rule introduced here: send the schema, never the rows
03	Prompt 2 (red flags)	Prompt 3 title says ICMJE, body asks CRediT	A planted-flag request with a model answer
04	none	'Stale `.Rhistory`' is not a common startup cause; a restored `.RData` is	Ask the agent to verify an install with `R --version`, `quarto check`
04b	Both	none	'Echo before rm' as a standing agent instruction
05	Prompt 3 (reflog first)	none	Commit before any agent session
05b	Prompt 3 (force push, revert)	Prompt 1's verification depends on the defective conflict example	none
06	Prompt 2 (failure handling)	Prices and AMIs from the model are stale by construction	Never paste or expose keys; budgets and alarms before an agent provisions anything
07	none	Prompt 1 must require a commit first and forbid moving `data/raw/`	none
08, 09	Layer-order checks	Pinned `rocker/r-ver:4.4.0` in prompts; no digest pinning	`.dockerignore`, digest pinning in the criteria
10	none	'zzcollab-compliant', `zzc analysis`, and the bundle schema are wrong against the current repository	Tool-neutral Five Pillars audit
10b	Both	none	none
11	Prompt 1 (inline numbers)	none	Criterion: change the data, re-render, numbers change

Chapter	Keep	Fix or replace	Add
12	none	Conversion prompts must not suggest `quarto convert` for Rmd	none
12b	All three	none	Treat scraped content as untrusted input to an agent
13	Prompt 1	Send the column names and a synthetic row, not five real rows	Agentic version on the running study
14	Prompt 3	Prompt 1 calls `parse_date_time()` with multiple orders 'canonical' without warning that ambiguous orders (dd/mm vs mm/dd) silently misparse; that is the chapter's own silent failure	none
15	Prompt 3	Add `unmatched = "error"` beside `relationship =`	Agentic version with the join checklist as criteria
15b	Prompt 1	none	Parameterized queries; never paste credentials or connection strings
16	Prompt 1	Prompt 2 steers away from `plot_annotation(tag_levels = "A")`, the idiomatic choice	none
17	Prompt 1	Prompt 3 should require named sensitivity analyses (delta, reference-based)	none
18	Prompt 2	Prompt 3 verifies against `zzpower`; use `power.prop.test`, `pwr`, or `rpact`, and note that the chapter's own SAP is underpowered	Estimand attributes as criteria
19	Prompt 2	Convert tip callout to the standard format	Independent re-derivation as the AI analogue of double programming
20	Prompt 1 (bug injection)	Name it mutation testing	Agents weakening tests; the tests-off-limits rule

Chapter	Keep	Fix or replace	Add
21	'Vibe coding' versus agentic coding; 'when not to'	`chat_claude()` is superseded by `chat_anthropic()`; `chat_async(stream = TRUE)` is not the ellmer API; `cor()` default is not pairwise; hard-coded model ID; CLI list omits Codex CLI, Gemini CLI, Positron Assistant	The four Cs; governance; agent safety; instruction files; `btw`/MCP; provenance and disclosure; move core to Part II
22	Prompt 1 (flag every NOTE)	Prompt 2 censoring sentence is garbled	none
23	Prompt 3	Prompt 2 proposes imputing sex 'from sex-specific distributions', which is circular	none
24	Prompt 1 (dropouts)	Prompt 3's '30% missing on a predictor: complete case is bad' contradicts the corrected theory in 17; DUA warning missing	Check the DUA before sending anything
25b, 25d	All	none	Use as the model for the rewrite
25c	Prompt 2	Prompt 3 steers toward dbGaP and TCIA, compounding the PHI-deposit error (Part 6)	none

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Part 5. Book-level restructuring

5.1 A running study

Create one synthetic study, distributed as a small data package or companion repository and generated by a seeded script, that every chapter from 01 to the capstone uses. The book already contains its pieces: the heart-failure PI in `03`, the readmissions cohort in 05 to 09, the HF-HOME trial in `25b`, the CRF extracts in `12b`, the dosing strings in `14`, and the ADSL/ADTTE build in `19`. Consolidate them into one study (provisionally HF-HOME). Its foundation is the single canonical SAP that Phase E1 produces (Finding 25, house rule 13): the current HF-HOME material contradicts itself across 03, 18, and 25b, and a running study built on it would spread those contradictions to every chapter. It would include the following.

- A messy CRF export with the planted file pathologies of 12b (BOM, leading zeros, mixed date formats, empty strings).
- Demographics, visits, labs (with a duplicate visit that breaks a naive join), dosing strings, and dropout.
- An SDTM-like layer and the ADaM build in 19, exported as `.xpt` for the SAS chapter.
- A small set of identifiers for de-identification exercises.
- A recurring cast: the PI, a co-analyst who causes the merge conflict, and a reviewer.

Penguins remains as the warm-up dataset where a non-clinical example reduces load (early ggplot2, the 23 capstone). ADNI remains the real-data case study, with an executable synthetic stand-in.

Two consequences follow. First, exercises become checkable, because everyone has the same data. Second, later chapters can say 'the join you wrote in chapter 15', which is retrieval practice at no extra cost.

5.2 Part order (adopted 2026-10-09, decision 3)

The current order puts infrastructure (Parts III and IV) before students have any data to protect. R4DS, Data Science in a Box, and Timbers and Cetinkaya-Rundel all argue for 'the whole game' first: early, visible results with real data, then the machinery. A revised order that also removes every forward dependency in Finding 18 is shown below. It is a recommendation, not a requirement. The template and the running study deliver most of the gain under either order.

Part	Chapters (current file names)	Change
Start here	Preface for students; 00-intro rebuilt as a 'whole game' tour: clone the study repo, render a report, change one line, watch a number change silently, catch it	Instructor material moves to an Instructor Guide appendix
I. Why and with whom	01-why; 03-team-science (trimmed); 02b concepts only	De-identification code moves after wrangling
II. Your toolkit	04b-shell, then 04-workstation (essentials only), 05-git-solo, 05b-git-teams (renv conflicts deferred); the durable core of 21-ai-coding if decision 11 is accepted	Shell before workstation; AI protocol before first use
III. Data in hand	12b split into two chapters; 13, 14, 15, 15b, 16, 17; 02b implementation section	Running study introduced in 12b
IV. Documents and environments	11 merged with the troubleshooting half of 12; 08-renv; 09-docker; 07 merged into 10 as 'Assembling the compendium'; 10b-targets; 06-cloud and HPC	All dependencies resolved in order
V. Analysis practice	18-analysis-plan plus the DMSP from 25c; 20-testing absorbing 20b-ci; 21-ai-coding (durable core, unless moved to Part II by decision 11)	Planning documents sit together
VI. Industry	19-cdisc, 22-sas (adjacent, sharing ADaM data)	Removes two independent ADTTE builds
VII. Analysis to audience	23-penguins (explicit map to chapters), 24-adni (executable), 25d-ethics (catalog), 25b-communicating	Ethics flags seeded earlier in their home chapters
VIII. Capstone	25-synthesis rebuilt as a capstone project with a rubric	See 5.4
Appendices	Cheat sheets; Tooling as of date; Instructor Guide; C, D, E, F	New appendices absorb perishable and reference material

The net change in chapter count is small. Three merges (07 into 10, 20b into 20, part of 12 into 11) offset one split (12b). The 25c deposition mechanics fold into the case studies.

5.3 Length targets

- 15 to 20 minutes of reading per chapter (3,000 to 4,000 words, including code).
- Chapters currently over 25 minutes are split or cut, with reference material moved out.
- Apply the cut list in Part 7 before adding anything. The target is a book about 15 percent shorter in words and longer in practice.

5.4 Capstone

Replace the recap with a project. Students take the running study from raw CRF export to a deposited compendium with a SAP, tests, CI, a one-page memo, and a reviewer response. The project has a rubric indexed to the numbered habits. `23-penguins` serves as the worked model, with each step labeled by the chapter that taught it. The chapter ends by answering the question posed in `01-why`.

5.5 Part introductions (reopens a Part 9 rule)

Each part of *Advanced R* opens with a page of about 400 to 500 words: one framing paragraph, a numbered one-line description of each chapter, and the advice that a reader who can answer a chapter's quiz may skip it. Part 9 of this plan forbids part-opener prose pages, on the grounds that `fig-book-map` already orients and that few readers proceed linearly. The *Advanced R* practice suggests the opposite inference from the same premise: a reader who enters mid-book is exactly the one who needs to know what each chapter assumes. A lighter version (100 to 200 words, no images, the question the part answers, one line per chapter, skip advice) is proposed as Part 11, decision 9. The image prohibition is unaffected.

5.6 A solutions companion

Advanced R Solutions (Grosser, Bumann, and Wickham) is a separate book whose numbering mirrors the main text, which keeps the main text lean. Immediate feedback is better supported for novices, so short keys stay adjacent to their exercises. The companion is for what has no home today: full model answers to the memo, the reviewer response, the DMSP, the SAP, and the capstone, with commentary on why each choice was made. It is also a natural contribution route for teaching assistants and former students (Part 11, decision 10).

5.7 Introduction and colophon

Two sections of the *Advanced R* introduction have no counterpart in `00-intro`, which already has 'What this book does not cover':

- **Getting help**: where to ask, and how to write a minimal reproducible example with `reprex`. Every first-year student needs this, and it is an instance of the book's own thesis.
- **Meta-techniques**: the few habits that apply everywhere. This is the natural home for the numbered habit list (4.3) and the motto (4.4).

Advanced R's colophon records the package versions used to build the book. The *Practicum* colophon describes fonts and tools and states that build reproducibility is managed through `renv` 'and a pinned `zzcollab` Docker image'. The repository has no Dockerfile (Finding 10), and the lockfile has never been restored, so that sentence is currently false (Part 6). Replace it with a statement generated at render time from the lockfile or `sessionInfo()`, which makes the colophon a demonstration of habit 2 rather than a claim.

5.8 Register model

Advanced R uses 'I' for the author, 'you' for the reader, and states limits plainly ('R is not a particularly fast programming language'). It is informal in grammar and precise in content. It is the concrete model for the middle course in Part 11, decision 1: the change is to hedging and indirection, not to precision or academic standards.

5.9 What not to borrow from *Advanced R*

- **Empty alt text on diagrams.** The *Practicum* is ahead here.
- **No end-of-chapter summary.** That is acceptable for intermediate readers; novices benefit from Rules of thumb and a cheat sheet (expertise reversal).
- **Solutions only in a separate book.** Immediate feedback should come first for this audience.
- **CC BY-NC-SA.** The *Practicum* deliberately chose CC BY so that it qualifies as an open educational resource.

Part 6. Correctness register

These are defects that a student would notice or reproduce. Fix them before any stylistic work, because each one costs trust. The register has four tables. The first holds defects that would cause harm or a wrong professional decision if acted on; fix them before anything else. The others are ordered by how they were established. Many entries in the third table were later confirmed independently by the expert reviewers; those are marked 'confirmed'.

Tier 1: fix first (expert reviews; harm or wrong decision if acted on)

Location	Defect	Evidence
25c:414-419, 180-183	Model DMSP deposits PHI-containing CRFs in dbGaP and PHI imaging in TCIA; both accept de-identified data only	inspected
02b:187-221, 258-286	Date shifting and MRN hashing labeled Safe Harbor; Safe Harbor permits year only; a re-identification code may not be derived from an identifier	inspected; hashing point expert knowledge
22:98-103	'FDA requires SAS-formatted deliverables'; FDA requires XPT transport files, not SAS	inspected
17:78-80, 168-172, 346-350; 23:216-222, 729-738	Complete-case analysis taught as biased under MAR; unbiased when missingness is independent of outcome given covariates; 'bounded by fraction lost' and the 5 percent rule unsupported	verified by simulation
17:886-914	`mice(seed = 1)` inside `replicate()` collapses 100 replicates to a few datasets; conclusions at 917-945 rest on it	verified

Location	Defect	Evidence
18:520-530, 288-292	Underpowered (1,094 per arm needed, not 750); attrition not applied (71, not 64); conditional OR called marginal; as-treated exposure; no estimand attributes; Bonferroni scheme lets FWER near 0.10	verified (power, attrition) and inspected
03, 18, 25b	HF-HOME inconsistent across chapters (size, power target, estimand, missingness); sensitivity code omits missing patients from denominator; reviewer response quotes 3.6 where code gives 3.8; immortal-time bias in the 03 intake	verified and inspected
20:541, 20b:177	`tinytest::test_all()` exits 0 on failure, so CI never fails	verified
PDF output (07:330, 11:156, 11:588-595, 12:615)	``` printed literally	verified with `pdftotext`
21 (whole chapter) and every LLM section	No guidance on PHI, DUA-restricted, or proprietary data sent to LLMs; 19:565 encourages pasting SDTM	verified by search
19:394, 420	`1L - CNSR` silently fails for CNSR other than 0 or 1; use `CNSR == 0`	verified

Verified or inspected directly by the plan author

Location	Defect
18:528-530	750 per arm for 80% power at p = 0.25 vs 0.20. `power.prop.test` gives 1,094 per arm; 750 gives 64% power (verified)
14:520-523	DST example: 2026 DST begins March 8, so `+ hours(24)` from March 9 crosses nothing. The frozen output prints `01:00:00` for both lines, contradicting the hand-written comment (inspected in `_freeze/`)
05b:339-371	'Preserve both intents' resolution is identical to the incoming side, since `sbp <= 220` subsumes `<= 250`; the later LLM check (05b:685-686) relies on a difference that does not exist (inspected)
17:169-171	Mechanism table marks complete case 'Biased' under MAR; the chapter's own Solution 2 shows near-unbiasedness under covariate-dependent MAR (inspected)
24:360	Two-year follow-up threshold used to classify a three-year outcome, the bias the chapter condemns at 79-88 (inspected)
all chapters	Verification, Pitfall, and LLM-prompt callouts promised but unused (verified)
colophon.qmd	States build reproducibility is managed by `renv` and 'a pinned `zzcollab` Docker image'; no Dockerfile exists and the lockfile has never been restored (inspected; render failure observed 2026-10-09)

Inspected by one reader, to confirm during editing

Location	Defect
04:367-369	Says `install.packages()` updates `renv.lock`; it does not without `snapshot()` (confirmed)
05b:460	`renv.lock merge=ours` needs a driver configuration and contradicts 'do not pick a side'; `--theirs` inverts under rebase (confirmed by test)
08:109-111, 143, 424-425	Snapshot `type` misdescribed; sandbox called 'the cache'; `::` detection misdescribed (confirmed)
09:220-221	`install.packages(..., version =)` has no such argument (confirmed by test)
07, 08, 09	`rmarkdown::render()` on `.qmd`; Dockerfiles disagree and do not render Quarto; `rocker/r-ver` called Debian (confirmed)
07:353-361	`rticles` presented as Quarto formats; NEJM template claimed; propensity score called 'the exposure' (confirmed)
11:411-422	`freeze` described as per-chunk and as the default (confirmed by test)
12:324-328, 423-427	knitr chunk headers stripped from examples (confirmed)
12b:485-486	`as.numeric('')` yields `NA`, not 0
13:164-174	Tidy-table example code references a column the table does not have
15:483-520	Worked example's final row-count assertion would fail on its own fixture
16:438-441 vs 532, 750	`theme_set()` called wrong, then used, then mandated
18:288-292, 526-528, 711	Attrition not applied; conditional OR called marginal; undefined `n_cluster_adj`
19:196-198, 322	SDTM dates described as ISO strings but built as `Date`
20:35 vs 937-945	Quiz question and answer name different functions
21:222, 635	`cor()` default is not pairwise; `filter` attributed to base
22:327, 362, 574, 727, 864	Nonexistent variable; 'five' introduces six; wrong reference level; MIXED/LOGISTIC label; garbled censoring sentence
23:148-155, 341, 735	Balance claim self-contradicts; hard-coded number after 'every number is computed'; MAR vs MCAR
24:389-424, 517	Object names never created; `vfold_cv(strata=)` used where grouped CV is needed
25b:146 vs 258	HF-HOME randomized 1,500 vs 2,706 patients; memo claims MI, code does not
25-synthesis	Habit map and habit list disagree; 'Two appendices' when there are four
C:77-79	`admiral` claimed to use `data.table` internally (likely false)
25b, 25d, D	British spellings ('per cent', 'fortnightly', 'catalogues') that the `us-spelling` hook misses

Additional defects from the expert reviews and the LLM-section review (evidence as labeled by the reviewer: V verified, I inspected, E expert knowledge)

Location	Defect
12:156-161, 241-247, 597, 736-739	`quarto convert` does not convert Rmd to qmd; it wrote a Jupyter notebook (V)
11:690	Exercise 4 on `freeze` cannot succeed on a single-file render (V)
20b	Workflow never installs Quarto; exercise names `publish.yml`, actual file is `render-book.yml` (V, E)
05:624, 463	Popped stash is not recoverable from the reflog (V); reflog expiry for unreachable entries is 30 days (I)
05:481	`echo "*.html\n*.pdf"` under bash writes one literal line (V); `.gitignore` omits `.Renviron`
10 (throughout)	Profiles, bundle schema, `make` targets, and `check-renv` behavior do not match the current zzcollab repository; zzcollab does not 'compose rrtools' (I)
04:448, 621; 06:213, 243; 09:296-301, 436	Public dotfiles as standard; long-lived IAM keys; RStudio on all interfaces with a literal password; committed database password (I, E)
06	`pip install awscli` is CLI v1; region-specific AMI; shebang on line 2 and no `trap`; `t3.medium` not free tier; EBS cost overstated (E, I)
04:160-165, 497, 265	No `quarto:` pin key (it is `quarto-required`); cask name `font-jetbrains-mono`; R installed twice; bash completion sourced in zsh (E)
10b:233, 304	`CMD` runs at container run, not image build; no `tar_option_set(packages =)`; no `tar_quarto()` (I)
C:124-129, 142	`dt[, x]` does return a vector (V); `melt(..., measure.vars = v1:v3)` errors (V)
14:286-291, 202-203	Factor joins match labels, not codes (V)
14:116, 445-460	`as.Date(45405)` is 2094, not 'year 124' (V); binomial intercept of 18.2 is an inverse-logit near 1 (V)
03:471-476, 487-490	p-value inconsistent with the stated interval (V); CI for a mean presented as an individual prediction (I)
03:534-566	Intake example has immortal-time bias, a causal verb, and 'Red flags noted: None' (I)
17:635-643, 389-391	CONSORT and STROBE item numbers misattributed; LOCF attribution to E9(R1) unconfirmed (E)
17:614, 617-663	Delta adjustment applied to both arms barely moves the contrast; regulators expect active-arm shift and a two-dimensional tipping point; edits `imp$imp` rather than using `post` (E, I)
17	Mean imputation and missing indicators for *baseline covariates in RCTs* are valid (White and Thompson 2005) but branded antipatterns (E)
18:113-120	Conditional HC1 is a pretest procedure; pre-specify HC3 (E)
19:75-80, 104-111, 361, 690	Roles (programmers build ADaM); mITT defined as safety set; analysis sets conflated with estimand population; `TRTEDT` from disposition; TRTEMFL incomplete (I, E)
22:462-484, 506, 551, 596-608	PROC MIXED example not equivalent to its R code; no MMRM; GENMOD default is PARAM=GLM; PHREG without CLASS errors; MERGE example implausible (V, I, E)

Location	Defect
24:356-372, 291, 429, 235	Early converters who drop out excluded; ADNIMERGE not on public GitHub; `theme = 'fda'` implies a nonexistent standard; APOE4 risk understated (I, E)
25c:174, 76-77	DMSP not scored in peer review; NIH 2024 Public Access Policy missing (E)
25d:295	'Roughly three times' is 2.2 times (V)
23:63-64, 226, 557-561, 745-752	iris baggage is eugenics, not colonialism (E); `na.omit` drops rows the model does not need (V); circular sex imputation (I); cross-sectional coefficient called an 'effect' (I)
25b:7-10	Whyte quotation misquoted (E)
21:124, 134, 525	`chat_claude()` superseded by `chat_anthropic()`; `chat_async(stream = TRUE)` not the ellmer API (V, ellmer 0.4.1); hard-coded model ID
14 LLM prompt 1	`parse_date_time()` with multiple orders recommended without the dd/mm vs mm/dd ambiguity warning (I)
16 LLM prompt 2, 16:724	Steers away from `plot_annotation(tag_levels = "A")` (I)
04 LLM prompt 2	'Stale `.Rhistory`' is not a common startup cause; restored `.RData` is (E)
24 LLM prompt 3	'Complete case is bad at 30 percent missing on a predictor' contradicts the corrected theory (I)

Part 7. Chapter actions

These are short directives that follow from the per-chapter readings. 'Cut' means move to an appendix or the Instructor Guide unless stated.

Chapter	Open with	Cut or move	Add	Merge or reorder
preface	Three silent failures	Peer-curricula section to Instructor Guide	'For students' page	Split students and instructors
00-intro	A reviewer asks for a rerun six months later	Eight-part prose (use table + map)	Whole-game tour on the running study	Remove duplication with preface
01-why	Duke/Potti or the three silent failures	Federal paragraph to a pointer	Break-it-then-fix-it script; failure-mode checklist	Name the motto here

Chapter	Open with	Cut or move	Add	Merge or reorder
02b	Sweeney re-identification	R implementation to Part III	Correct Safe Harbor versus Expert Determination labeling (Tier 1); mnemonic for suppress/generalize/hash/shift; Limited Data Set with DUA; LLM data-governance rule	Concepts stay in Part I
03	The heart-failure PI's first email	Duplicate disagreement section; results communication to 25b	Scenario exercises with model answers	Recurring PI across chapters
04	Keep current hook	Fonts, tmux, starship to 'Level up' appendix	Per-OS tabs including WSL; 'Is my machine ready?' checklist	After 04b
04b	A file with a planted BOM	None	Shipped `visits.csv`; solutions; shell-vs-R table	Before 04
05	March vs November analyst	Two of three 'why' passes; `zzgit`	'Oh no, I…' recovery table; conflict and detached-HEAD sections	None
05b	Two analysts, 1,847 vs 1,912 rows	Meta callout; renv.lock conflicts to 08	Fixed conflict example	Second person throughout
06	A job that ran out of memory	AWS catalog to glossary box; prices to dated table	Minimal SLURM `sbatch` example	After 09
07 + 10	A collaborator gets a different p-value	rticles; zzcollab bundles and extension	Tool-neutral Five Pillars checklist; one canonical Dockerfile; any remaining zzcollab text rewritten against the current repository	Merge as 'Assembling the compendium'
08	Keep silent-regression hook	Dockerfile section to 09	renv verbs card; P3M snapshots	None
09	Same code, different number	Compose to a pointer	Apple Silicon first; CI-tested Dockerfile	None
10b	Table 2 from a stale intermediate	Peer-survey callout	Runnable pipeline with real `tar_make()` output; `tar_quarto()`	After 11

Chapter	Open with	Cut or move	Add	Merge or reorder
11 + 12	A number typed into a sentence	YAML dump and website to reference	Runnable mini-project; freeze vs cache table; error/cause/fix table	Merge; retitle remainder 'Troubleshooting documents'
12b	Keep current hook	Duplicate callout	Import checklist card	Split in two; owns the raw-data rule
13	The running study's CRF export	`stringsAsFactors`, `%>%` digressions; pandas to an aside	Exclusion counts at each filter; verbs card	Continues from 12b
14	Keep current hook	API tours to cheat sheet	Problem-then-verb vignettes on dosing data	None
15	Keep current hook	`rows_*` to an aside	Visible fixtures; join checklist reused in 15b and 18	None
15b	Credentials, not a CSV	Parity callout	SQL-to-dplyr card; same tables as 15	None
16	Before/after makeover of a bar-of-means figure	DPI folklore; `zzlongplot` beside the portable alternative	Figure checklist	Trajectory figure as spine
17	A trial with dropout	LOD aside and reporting detail to 'going further'	Corrected complete-case theory and rerun simulation (Tier 1); mechanisms by story before notation; MMRM, reference-based MI, tipping point (Phase K)	Defer pre-specification to 18
18	An outcome switch caught by audit	Duplicated Orientation	Canonical HF-HOME SAP with estimand attributes, corrected power, named multiplicity procedure (Tier 1); downloadable SAP skeleton; DMSP from 25c; Gamble et al. (2017) in place of '243B'	Planning documents together
19	Keep alumni hook, moved to top	Four of five CNSR restatements; Sources	ADaM variable card; define estimand	Adjacent to 22
20 + 20b	The 86,400-fold drift	testthat comparison to the end; duplicate worked example	Red CI log excerpt; offline exercises with keys	Merge

Chapter	Open with	Cut or move	Add	Merge or reorder
21	Keep thesis	ellmer and model strings to dated appendix; styler/lintr earlier	Executed planted-bug examples with assertions; the four Cs, governance, and agent safety (4.6)	Core to Part II if decision 11 is accepted; otherwise after 20
22	Screening-stage exclusion	ODA click-path	Correct 'FDA requires SAS' (Tier 1); defaults-that-differ table; offline log-reading exercises; MMRM in both languages; R submission pilots	Reuse 19's ADaM
23	Keep framing	Literal `paper.qmd`	Step-to-chapter map; testing and CI steps	Worked model for capstone
24	A family asking 'will this progress?'	'243B'	Executable synthetic run; `gtsummary` alternatives	None
25d	OkCupid scrape	Meta callout	Solutions; inline statistics	Ethics flags seeded in 02b, 12b, 16, 25b
25b	'So did it work?' email	Meta callout	Before/after rewrite of a chronological memo	Final act before capstone
25c	DMSP due before the work begins	Dated policy to an 'as of' table; renv repetition	Filled DMSP for the study with de-identified deposit and appropriate repositories (Tier 1)	DMSP to 18; deposition to case studies
25	The question from 01-why	Recap and reading list	Capstone project with rubric; reconciled habit list	None
F	n/a	None	Links from every bold term; missing terms; A to Z order	None

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Part 8. Phased plan

Effort figures are estimates in working days for one person, offered for sequencing rather than for scheduling, and they assume AI assistance for the mechanical parts with human verification of every result.

Phase A. Adoptability unblock (about 2 days)

Small, self-contained, and a precondition for everything in Phase D. Item 1 is now a correction of a published error rather than an improvement, which raises its priority above everything else in this plan.

1. Relicense prose to CC BY, keep code under MIT or GPL-3, and make `LICENSE.md`, `analysis/report/index.qmd`, `DESCRIPTION`, `CITATION.cff`, and the colophon agree with each other. The published book currently tells readers it is CC BY-NC-ND.
2. Set `toc: true` under the `pdf` format in `_quarto.yml` and confirm the rendered PDF carries a usable table of contents. One line.
3. Fix the two open items from `REVIEW-CHECKLIST.md` on `00-intro`: the appendix count and the missing 'Sources' entry in the chapter pattern list.
4. Move each chapter's `Prerequisites answers` section adjacent to its quiz, or make it reachable by an anchor link from the quiz, rather than leaving it as the last heading of the chapter.
5. Add a `fig-alt` to every figure that currently exists.

Acceptance. All five license declarations agree and none contains ND or NC for the prose; the rendered PDF opens on a table of contents; `00-intro` describes three appendices; no chapter's answer section is its last heading without a link from the question; every figure that renders carries a `fig-alt`.

Status: complete as of 2026-09-07 18:40 PDT, except for the render verification. Item 1 done across `LICENSE.md`, `analysis/report/index.qmd`, `colophon.qmd`, `DESCRIPTION`, and `CITATION.cff`; a fifth disagreement was found and resolved in passing, since `LICENSE.md` had called printed code MIT while `index.qmd` called it CC0. Item 2 done (`toc: true`, `toc-depth: 2`). Item 3 done. Item 4 required no work: all 34 chapters already carried `@sec-quiz-answers-NN` from the quiz. Item 5 done, 43 alt texts, which is every figure that renders. The five remaining `fig-` labels sit inside display fences and produce no image, so they take no alt text and are a Finding 2 item instead.

Not yet verified. The book has not been rendered since these changes. `toc: true` under `scrbook` is untested, and the alt text has not been seen in built output.

Phase B. Practice with feedback (about 8 to 12 days)

The highest-value phase. Do it chapter by chapter in curriculum order.

1. Per chapter, classify existing exercises as open-ended or checkable, and label the open-ended ones explicitly.
2. Per chapter, add three to five checkable exercises built on a dataset the book already ships.
3. Write worked solutions for every checkable exercise and place them adjacent to the question, in a collapsible callout for HTML and a per-chapter solutions block for PDF.
4. Rebalance exercise counts to roughly one per major section.
5. Add a retrieval prompt per major section, with the answer immediately available.

Acceptance, as originally written. Every chapter has at least three exercises with published solutions; no chapter has exactly five exercises by coincidence of template; a reader with no project of their own can self-assess in every chapter.

Status: 2026-09-08. The third criterion is not achievable and should be struck. Fourteen chapters now carry worked solutions (45 executable solution chunks): `02b-deidentification`, `12b-acquisition`, `13-wrangling`, `14-types`, `15-joins`, `15b-databases`, `16-graphics`, `17-missing`, `18-analysis-plan`, `19-cdisc`, `20-testing`, `23-penguins`, `24-adni`, `25d-ethics`. Every one of the remaining twenty chapters has had its exercise set labeled as open-ended, with a one-line statement of *why* no answer key exists.

The limitation this plan anticipated turned out to be structural rather than incidental. The twenty labeled chapters are the infrastructure and practice half of the book, and their exercises are not checkable in principle: they modify a machine's state (`04-workstation`, `08-renv`, `09-docker`, `07-rrtools`, `10-zzcollab`), require a collaborator and a remote (`05b-git-teams`, `20b-ci`), need software the book does not ship (`22-sas` needs SAS; `21-ai-coding` needs API credentials), or produce a document rather than a value (`25b-communicating`, `25c-federal`, `01-why`, `03-team-science`).

Inventing checkable exercises for those chapters was considered and rejected. An artificial exercise on `palmerpenguins` in the Docker chapter would satisfy the acceptance criterion and teach nothing about Docker. `20-testing` was the one case where new exercises were genuinely warranted, and three were written.

So the honest target is: **every chapter either publishes solutions or states why it cannot**, which is now met. A reader with no project of their own can self-assess in fourteen chapters and is told plainly where they cannot.

Phase C. Executable code and figures (in progress since 2026-09-07 19:00 PDT)

Progress as of 2026-09-08. Executable `{r}` chunks: 48 to 118. Top-level ```` ```r ```` display blocks: 128 to 60. Sixty-eight blocks converted across fourteen chapters. The full book renders clean with no warnings.

The remaining 60 are mostly not convertible, and the reason is structural rather than incidental. The chapters left are the infrastructure ones, where the code's purpose is to change the state of a machine: `08-renv` (9 blocks, all `renv::init`, `snapshot`, `restore`, `install.packages`), `07-rrtools` (compendium scaffolding and `use_github`), `04-workstation` (`.libPaths()` output for a hypothetical machine, and an `.Rprofile` listing), `10b-targets` (`tar_make()` would run a pipeline), `24-adni` (needs the restricted ADNIMERGE package), `21-ai-coding` (live API calls), `15b-databases` (a remote Postgres connection), and `12-rmd-workflow` (one block sets `opts_chunk` globally, which would leak into the rest of the render).

Running these would mutate the project's own lockfile and library, publish repositories, or reach the network at build time. They are correctly display-only and should receive the dated verification note of item 3 rather than conversion.

Revised expectation. The realistic ceiling is roughly 75 to 80 conversions of the original 128, not the 115 estimated at the higher bar on 2026-09-07. That estimate was extrapolated from data-analysis chapters and did not account for the infrastructure half of the book, where display-only is the correct state for nearly every block.

A blocking build failure was fixed first, and it was not in the plan. The book did not render at all: `13-wrangling` chunk `wrangle-pandas` died on `ModuleNotFoundError: No module named 'pandas'`. The `Makefile` documents this ('a build without deps-py fails at 13-wrangling.qmd') and expects `pip install -r requirements.txt` against the ambient interpreter. That is precisely the dependency-on-machine-state this book argues against everywhere else. It is fixed by declaring the requirement in the document instead: `reticulate::py_require()` in the four chapters with Python chunks (`13-wrangling`, `14-types`, `15-joins`, `16-graphics`), so reticulate provisions its own environment through uv. `requirements.txt` and the `deps-py` target are now redundant for the book build and should be reviewed.

The 10-to-15-day figure was estimated before Finding 3 was given a countable target. Bringing 170 code-bearing sections to a rendered artifact from a base of about 20 is a larger job than that estimate covers, and the ambient images add sourcing and attribution work that is not development time at all. Treat this phase as the one most likely to overrun, and re-estimate after the first three chapters.

1. Triage all 258 display-only blocks into executable, not executable, and expensive. Start with the heaviest files: `22-sas` (25), `11-quarto` (22), `12-rmd-workflow` (19), `05b-git-teams` and `13-wrangling` (17 each).
2. Convert the executable subset to `{r}` chunks, one chapter at a time, rendering after each chapter.
3. ~~For blocks that cannot execute, add a dated verification note stating how and when the command was checked.~~ **Revised 2026-09-08.** Replaced with an automated check, for two reasons.

   A dated note is a claim about the past that decays from the moment it is written, and 60 of them across the book would be 60 assertions nobody re-checks. Worse, most could not be written honestly in the first place: verifying `renv::init()`, a `docker build`, or an AWS provisioning sequence means running it against a real project, daemon or account, and writing 'verified' beside a command nobody ran is the failure mode this whole plan exists to correct.

   What *is* checkable, mechanically and repeatedly, is whether the R functions those blocks name still exist. That is the part of the code-rot exposure a rendered build cannot see, because `freeze` protects only executed chunks.

   The book already had this check, and it was broken in two ways. `tools/check-calls.R` predates this plan, is wired into `make check`, and its header records that it once caught five nonexistent calls the

render never would have. I wrote a duplicate before looking in `tools/`, which was wasted work and a reminder to read the repository before adding to it. The duplicate is deleted; both bugs below are fixed in the original.

It verified nothing when run the way the Makefile runs it. The script called `list.files(".")`, so from the repository root it found no `.qmd` files, reported `ok: 0 resolvable calls verified`, and exited 0. A passing check that checks nothing is the vacuous test `20-testing` warns against, sitting in the project's own toolchain. It now resolves `analysis/report` explicitly and verifies **106 calls, 18 unverifiable here**.

It reported a false failure when run correctly. `haven::write_xpt` was flagged as not exported. The call is real; the extracted token was `write_xpt.` with a sentence-ending period absorbed, because an R function name may legitimately contain a period and the pattern cannot tell punctuation from identifier. Trailing periods are now stripped before resolution. The offending sentence was one I added to a `fig-alt` in `22-sas` earlier in this same session, so the check caught a defect introduced by the refactoring itself.

Note what none of this covers: shell, Docker, cloud and SAS commands remain unverifiable without a human running them, and no note in the text should imply otherwise. Writing 'verified' beside a command nobody ran is the failure mode this plan exists to correct.

4. Bring every code-bearing section to at least one rendered artifact, which is 170 sections against the current ~20, prioritizing the five files that currently have no figure of any kind: `11-quarto`, `25b-communicating`, `C-alternative-wrangling`, `D-peer-survey`, and `E-lab-software`, then `12-rmd-workflow`, `20-testing`, `24-adni`, and `17-missing`. Leave the 204 prose-only sections alone; do not invent figures to fill them.

5. Handle `25b-communicating` as the stated exception, adding code so that it acquires three or four rendered artifacts rather than being exempted for having none.

6. Source and place the five portraits named in the April image policy, complete their `credits.qmd` rows and the cover attribution, and add no other ambient imagery.

7. Add `fig-alt` to each new technical figure as it is created; give ambient images a null `alt` if they are genuinely decorative, and reclassify them as technical if they are not.

Acceptance. A full render completes clean; no content file has zero figures; every code-bearing section produces a rendered artifact in the built book, verified by rendering rather than by grep; ambient images number no more than about twenty and all carry provenance.

Phase D. Reference apparatus and distribution (about 5 days)

1. Build the glossary appendix by extracting bolded first-use terms.
2. Add a PDF index.
3. Do not split chapters on length. Superseded by the rendered page counts; see Finding 8.
4. State the modular-assignment policy in the preface.
5. List in the Open Textbook Library and solicit reviews.
6. Produce a print-on-demand paperback.

Acceptance. Glossary and index present in both output formats; no chapter over about nine estimated pages; catalog listing submitted.

Status: 2026-09-08. Two of six done; four are blocked on decisions or actions only the author can take.

Done. The glossary is written and shipping as appendix F (`F-glossary.qmd`), roughly 75 terms across ten domains, each cross-referenced to the chapter that develops it. The modular-assignment policy is stated in the preface under 'Assigning parts of this book', including the honest caveat that Parts I to III are a dependency chain rather than independent modules.

Not done, and why.

The index needs a decision before it needs work. A back-of-book index requires `\index{}` markup at every occurrence of every term across 38 chapters. That is not a scan-and-insert job: the markup is LaTeX, so it has to be confined to PDF output or it leaks into HTML as literal text, and deciding which occurrences of a term deserve an entry is editorial judgment, not pattern matching. Note also that the HTML edition already has full-text search and the PDF now has a table of contents, so the index is a PDF-only gain. Recommend deciding whether it is worth the source clutter before anyone starts.

The five portraits are blocked on licensing, which should not be automated. The April image policy names Bryan, Marwick, Allaire, Xie and Wickham, requires CC-BY or CC0 sourcing, and requires a `credits.qmd` entry per image. Sourcing and verifying the license of a photograph of a living person is a judgment with real consequences if it is wrong, and it is not a task to delegate to a tool that cannot see the license page it is reading. `images/portraits/` is still empty and five `credits.qmd` rows still read TODO, along with the cover's own attribution line. (*Superseded later on 2026-09-08 by commit `57b9bc5`: portraits of Gentleman and Wickham were placed and the reasons the others are absent were recorded. Gentleman is not on the April policy's list of five; the policy or the placement should be reconciled. Three portraits remain open; see Part 11, decision 8.*)

Chapter splitting was not attempted. Six files exceed 4,000 words. Splitting them is an editorial act that changes the book's structure and its cross-references, and it should follow from a reading rather than from a word count.

Catalog listing and print-on-demand are the author's to perform. Both require accounts, identity, and a submission the author stands behind. They also depend on Phase B being far enough along that a reviewer would not immediately note the missing solutions, which is now true for fourteen chapters and not for the rest.

Finding 10 follow-up (status 2026-10-09)

Item 1 is done in form only: `renv.lock` was added on 2026-09-09 (commits `1459c73`, `f17ba8d`) and records 215 packages. It has never been restored. On 2026-10-09 a PDF render failed at `12-rmd-workflow.qmd` because the project library was empty (`knitr` and `rmarkdown` not installed), and the book rendered only with `RENV_CONFIG_AUTOLOADER_ENABLED=FALSE` against the global library. The lockfile is therefore untested, and the numbers in the current PDF come from whatever versions happen to be installed globally. Items 2 to 4 (`.Rprofile`, executing CI, Dockerfile) are open. Run `renv::restore()` once, render, and fix whatever fails; until then the book does not meet its own second pillar in practice.

Phases E to K. Correctness, readability, and content (from the 2026-10-09 reviews)

Effort estimates are approximate and assume one author working with assistance. Phase E is independent of every decision in Part 11 and should start first.

Phase E. Correctness (about 9 to 14 days; revised upward after the expert reviews). Two stages.

E1, Tier 1 (about 5 to 8 days). Fix every item in the Tier 1 table of Part 6 before any other work, and before the book is assigned again. The largest item is a single canonical HF-HOME SAP (estimand attributes, randomized assignment, corrected power and attrition, a named multiplicity procedure), from which chapters 03, 18, and 25b compute every number they quote. Rewrite the missing-data theory in 17 and 23 around the condition under which complete-case analysis is unbiased, and rerun the simulation without the inner seed. Correct the compliance statements in 02b, 22, and 25c, and have a privacy officer or regulatory colleague read the corrected 02b and 25c. Fix the CI exit status. Add a render-time check that fails the build if ``` appears in output.

E2, the rest of Part 6 (about 4 to 6 days). Fix the remaining defects in Part 6, confirming each 'E' item against its primary source first. Make the remaining display-only worked examples that *can* execute do so, in particular the worked examples in 15, 17, and 24 (the last against a synthetic stand-in for ADNIMERGE, since the real data are restricted). Replace every hand-typed `#>` line in an executed chunk with real output. Build and CI-test one canonical Dockerfile and use it verbatim in 07, 08, and 09. This phase also closes the Finding 10 follow-up above.

Note the tension with Phase C. Phase C correctly concluded that most blocks in 10, 10b, 11, and 12 should stay display-only because running them would mutate the project or reach the network. Phase E does not reverse

that. It proposes a small, self-contained example project inside each of those chapters (a toy `_targets.R`, a toy Quarto project in a temporary directory) that can execute without touching the book's own state, so that the reader sees real output.

Phase F. Conventions and template (about 3 days). Rewrite `conventions.qmd` to the four-callout set of Part 4 and set `reference-location: margin` for footnoted asides (4.1). Write the numbered habit list and the motto, and draw the recurring pipeline diagram (4.5). Move all metadata callouts (Sources, peer-survey justifications, curriculum parity) into an Instructor Guide appendix. Prototype the template on two chapters of different kinds, for example `15-joins` and `08-renv`, and pilot them with two or three students before continuing.

Phase G. Running study (about 5 to 8 days). Write the seeded generator and companion repository described in Part 5. Rewrite exercises in 04b, 12b, 13, 14, 15, 15b, 16, 17, 19, 20, and 22 against it, with adjacent keys. This also revisits a Phase B judgment: several chapters labeled open-ended in Phase B (04b, 05, 05b, 22, 25b) admit checkable exercises once shared data and a shared repository exist, for example a planted conflict in the study repository, or SAS log reading that needs no SAS installation.

Phase H. Openings, register, and cuts (about 1 to 2 days per chapter, 25 to 40 days total). Apply the template, opening story, retrieval checks, cut list, and the Finding 31 table and figure candidates chapter by chapter, most-assigned chapters first (13, 15, 16, 17, 01, 05, 08, 11, 18, 20). The 30 candidates marked 'Ready' in Finding 31 do not depend on any phase and may be done earlier as quick wins. The register change in this phase follows decision 1 in Part 11 (the middle course: remove hedges, define acronyms, second person in instructions, exercises, and callouts).

Phase I. Reorder and capstone (about 5 days). Apply the part order if adopted (Part 11, decision 3), resolve the remaining forward references, rebuild 25 as a capstone, and seed the ethics flags in their home chapters.

Phase J. Reference apparatus (about 3 days). Cheat-sheet appendix, 'Tooling as of' appendix, glossary links from bold first uses, 'Getting help' and 'Meta-techniques' sections in `00-intro`, a generated colophon (5.7), part introductions (decision 9), and a `testthat` translation appendix (decision 14). Webr or Quarto Live for the closed exercises in the data chapters is optional and has consensus support only. This phase absorbs the open items of Phase D (no index, per decision 8; portraits are the author's).

Phase K. Content gaps (about 10 to 15 days; scope depends on decisions 12 to 14). The expert reviews (Finding 28) list topics a professional reader expects. Consistent with the rule against adding chapters (Part 9), these enter as sections of existing chapters, prioritized by what an entry-level hire needs first:

1. *LLM governance and agentic practice* (21, moved to Part II per decision 11; see 4.6).
2. *Estimands* in 18 (the five ICH E9(R1) attributes, intercurrent events, a worked HF-HOME estimand), cited from 17 and 19.
3. *Missing data for trials* in 17: MMRM, reference-based multiple imputation, tipping-point analysis; the NRC (2010) report.
4. *Security basics* across 04, 06, 09, 15b: secrets and `keyring`, SSO instead of keys, parameterized SQL, `.dockerignore`, binding to localhost.
5. *Reproducibility cases and scholarship* in 01: Duke and Potti, Reinhart-Rogoff, NASEM (2019).
6. *Industry practice* in 19 and 22: pharmaverse TLF tools, the R submission pilots, validation of R packages, double programming and `diffdf`, an MMRM in both languages.
7. *Currency* in 04, 08, 09, 20b: Posit Package Manager snapshots, devcontainers, pre-commit, DuckDB and Parquet, pinned Actions.
8. *Causal foundations* where the book already uses causal language (23, 25d): a DAG, target-trial emulation, and the 'C-word' argument, in a page each.
9. *Comparator texts* cited in the preface and in Further reading.

To keep the book from growing, pair every addition with a cut from the same chapter (Part 7 and the R4DS overlap in decision 13).

Pilot loop. After each phase, collect a short reading-time and confusion log from two or three students. The Carpentries require pilots for good reason: 'you are not your learners' (Wilson).

Part 9. What not to do

The first two items below were written on 2026-09-07 and are retained with their original reasoning; the October reading contested both, and both were resolved on 2026-10-09 (Part 11, decisions 1 and 2).

- ~~Do not rewrite prose for voice.~~ **Resolved 2026-10-09, decision 1: the voice is kept; hedges are removed, acronyms defined, and instructions, exercises, and callouts address the reader as 'you'.** The September reasoning was that the rgtvoice pass is complete, `DEEP-REVIEW.md` found the register consistent, and further prose churn would consume the budget that Phase B needed. The October reading agrees the register is consistent and finds that it is consistently above the audience (Finding 15); the middle course answers both.
- ~~Do not restructure the book to reach 140 pages.~~ **Retained in substance; resolved 2026-10-09, decision 2.** No restructuring toward a page target is proposed. The October plan proposes moving reference and perishable material out of tutorial chapters, which shortens them, and splitting one chapter (`12b-acquisition`) on content grounds rather than length.
- Do not delete the mermaid diagrams in favor of computed figures. They serve a different purpose and reviewers do not complain about schematics; they complain about their absence of everything else.
- Do not retain the NC or ND terms, and do not adopt them elsewhere to retain commercial control. (Resolved: the prose is CC BY 4.0.)
- Do not add humor, trivia, or decorative photographs to chase engagement. The seductive-details literature shows harm (meta-analytic g of about -0.33), and the book's `images/README.md` states the rule more strictly: no chapter-opener images, no generated hero images, no part-divider art, and four to five portraits at most. Stories are kept only when they are instances of the chapter's failure mode.
- Do not add part-opener images. ~~Or prose pages.~~ **Prose pages permitted from 2026-10-09 (5.5; Part 11, decision 9): 100 to 200 words, image-free.** `fig-book-map` in `00-intro` orients a linear reader once; a short part introduction serves the reader who enters mid-book.
- Do not set a uniform per-section figure count. A figure invented to satisfy a quota in a prose section is a seductive detail with a `ggplot` theme.
- Do not add chapters. The problem is density and delivery, not coverage. The content gaps the expert reviews identified (Phase K) enter as sections of existing chapters, each paired with a cut; decision 12 is the one possible exception.
- Do not convert prose wholesale into bullet lists. Scannability comes from headings and short paragraphs; the book's best passages are prose and should stay prose.
- Do not let the LLM prompts grow back into sections. One callout, beside the task.
- Do not remove the bypass quiz without replacing it. Retrieval with feedback is the strongest effect available; it moves and multiplies, it does not disappear.
- Do not let in-house tools carry concepts. Every `zz*` mention needs its portable alternative within the same paragraph.

Part 10. Sequencing

Phases A to D were sequenced as A, B, C, D, on the grounds that A was short and unblocked D, B carried the most evidential support per unit of work, C generated figures as a byproduct, and D (catalog listing) should wait until exercises had solutions. A is complete, B and C are substantially complete, and D's remaining items were settled by decision 8 on 2026-10-09.

The remaining work should run in this order:

1. **Phase E1 (Tier 1 defects).** Before the book is assigned again. Three expert reviewers independently judged the book not adoptable until these are fixed.
2. **Phase E2 and the Finding 10 follow-up.** Independent of every decision, and every defect in Part 6 costs reader trust.
3. **Part 11 decisions.** Recorded 2026-10-09; only item 15 (the portrait policy) remains open, and it blocks nothing.

4. **Phase F, then G.** The template and the running study are the two changes with the largest effect per unit of work. If only one readability phase is ever done, do Phase G: shared data makes exercises checkable, gives the book a through-line, and supplies spaced retrieval at no additional cost. Phase G depends on the canonical SAP from E1.
5. **Phases H, K, I, J,** chapter by chapter, with the pilot loop. Phase K's additions to a chapter are made in the same pass as that chapter's Phase H rewrite, so each chapter is opened once.
6. **Phase D's catalog listing and print-on-demand** last, for the reason the September plan gave, and not before an external expert has re-read the corrected statistical and compliance chapters.

Part 11. Decisions for the author

Decisions recorded 2026-10-09

The author accepted the plan's recommendation on every decision. The options and arguments that informed each are retained below the table as the record of what was weighed.

#	Decision	Resolution
1	Register	**Middle course.** Keep the author's voice; remove hedges ('we would argue', 'it appears'); define every acronym at first use in each chapter; use second person in instructions, exercises, and callouts. No book-wide rewrite to second person.
2	Chapter splits	**Split `12b-acquisition` only**, at its seam between files and APIs or scraping. Elsewhere, shorten by moving reference and perishable material out (04, 06, 11, 15, 16, 17, 21), not by splitting.
3	Part order	**Data-first order of 5.2**, adopted together with the running study (decision 4); the two are a package.
4	Running study	**HF-HOME**, built on the single canonical SAP produced in Phase E1.
5	In-house tools	**Optional, community tool first.** Teach `power.prop.test`/`pwr`/`rpact`, `gtsummary`, `ggplot2`, and `renv` with Docker as primary; present each `zz*` package as an option with its validation status. zzcollab becomes one section of a tool-neutral 'Assembling the compendium' chapter (07 merged with 10), rewritten against the current repository.
6	Interactivity	**Deferred.** No webR or Quarto Live for now; revisit after the pilot loop.

#	Decision	Resolution
7	Instructor material	**Appendix** in the book (an Instructor Guide), for ease of keeping it in sync.
8	Phase D carry-overs	**No PDF index** (HTML search and the PDF table of contents suffice). Remaining portraits: the author sources and license-checks them personally. Catalog listing and print-on-demand: **after** human expert re-review of the corrected statistical and compliance chapters.
9	Part introductions	**Yes**, 100 to 200 words each, image-free (5.5). The Part 9 rule against part-opener prose pages is withdrawn; the rule against part-opener images stands.
10	Solutions companion	**Yes, conditionally**: start it if a teaching assistant or former student will co-author. Otherwise, collapsed keys adjacent to exercises suffice.
11	AI chapter placement	**Move the durable core of 21 into Part II** (the four Cs, governance, agent safety); tooling detail goes to the dated 'Tooling as of' appendix.
12	Estimands	**A section of 18**, not a new chapter. Revisit if it exceeds about 1,500 words.
13	R4DS overlap	**Cut** the generic verb tours in 13 to 16 and the pandas side-by-sides; point to R4DS 2e; keep the clinical-judgment layer. This offsets the Phase K additions.
14	Test framework	**Keep `tinytest`**, and add a short `testthat` translation appendix (the existing testthat comparison callout in 20 moves there).
15	Portrait policy (new)	**Open item for the author**: the Gentleman portrait is placed but Gentleman is not among the five named in the April image policy (`images/README.md`). Amend the policy or replace the portrait.

Consequences for the rest of this plan: house rule 2 and Phase H apply the middle course; Part 5.2 is adopted rather than proposed; Phase K proceeds with estimands in 18; Phase J omits the index and includes part introductions; the remaining `[conditional on decision N]` wording elsewhere in this document should be read in light of the resolutions above.

Options as weighed

1. **Register (reverses a September position).** Keep the rgtvoice register, or move to second person for this audience?
   - *For changing:* a meta-analysis of conversational style reports retention d = 0.30 and transfer d = 0.54 (Ginns, Martin and Marsh 2013), and the book's most memorable lines already use direct address. Formal hedging ('we would urge the reader') is a cost a first-year student pays on every page.
   - *For keeping:* the voice is deliberate and consistent; the effect sizes are modest and some later studies find smaller benefits; the effect fades in lessons over about 35 minutes; and a full pass costs weeks.
   - *Middle course:* keep the voice, but remove hedges ('it appears', 'we would argue'), define every acronym on first use, and put second-person address in instructions, exercises, and callouts only.
2. **Chapter splits (reverses a September position in part).** The September plan split nothing, on rendered page counts. The October reading proposes splitting `12b-acquisition` (the longest reading unit, about 30 minutes, with two separable halves) and moving reference material out of 04, 06, 11, 15, 16, 17, and 21, which shortens chapters without splitting them. Accept, or keep the current units?
3. **Part order.** Adopt the data-first order of Part 5, or keep infrastructure first and apply only the dependency fixes of Finding 18? The template and the running study work under either.
4. **The running study.** Consolidate on HF-HOME, or another domain closer to the program's applied focus?
5. **In-house tools (extended by Finding 29).** zzcollab: one section in a tool-neutral compendium chapter, keeping the Five Pillars as the durable idea, or a full chapter as now (rewritten against the current repository either way)? All `zz*` packages: teach the community tool first (`power.prop.test`, `pwr`, `rpact`; `gtsummary`; `ggplot2`; `renv` with Docker) and present the in-house package as an option with its validation status, or keep the current positioning? All three expert reviewers recommended the former.
6. **Interactivity.** Is webR or Quarto Live worth the build and maintenance cost? No outcome evidence specific to the format was found.
7. **Instructor material.** An appendix in the book or a separate instructor site?
8. **Carried over from Phase D.** Whether a PDF index is worth the `\index{}` markup; sourcing the three remaining portraits; catalog listing and print-on-demand (now a 712-page product).
9. **Part introductions (reverses a Part 9 rule in part).** Add a 100-to-200-word, image-free introduction to each part, as in *Advanced R* (5.5), or keep `fig-book-map` as the only orienting device?
10. **Solutions companion.** Start a separate model-answers volume for the long-form deliverables (5.6), and if so, invite teaching assistants or former students to co-author it?
11. **Placement of the AI chapter.** Move the durable core of `21-ai-coding` (the four Cs, governance, agent safety) into Part II so it precedes the 30 chapters whose LLM callouts depend on it, leaving tooling detail in a dated appendix (4.6)? Students use these tools from the first week regardless of where the chapter sits.
12. **Estimands.** Treat ICH E9(R1) estimands as a section of 18 (Phase K), or as the one new chapter the plan would admit? The pharma reviewer listed it first among absent topics.
13. **Overlap with R4DS 2e.** Cut the generic verb tours in 13 to 16 (and the pandas side-by-sides) and point to R4DS, keeping only the clinical-judgment layer, as the academic reviewer recommends? This is also the largest available source of length to offset Phase K.
14. **Test framework.** The book teaches `tinytest`, partly because zzcollab ships it. The data-science reviewer argues that `testthat` fluency matters more for a first job. Keep `tinytest`, switch, or teach `tinytest` with a short `testthat` translation appendix?

Part 12. Limitations of this analysis

Structural pass (2026-09-07 to 2026-09-08)

This is a mechanical scan, not a reading. Word counts include code blocks and YAML, so page estimates run high; the 450-words-per-page divisor is conventional rather than measured against this book's rendered output. Exercise counts come from counting numbered list items under an `## Exercises` heading and will miss

exercises formatted differently. The figure counts distinguish mermaid diagrams from R-generated figures by chunk syntax and would misclassify a figure included by another route, though no markdown image includes were found in any chapter.

I have not assessed cultural relevance, which is the second-weakest criterion across the Open Textbook Library corpus at 66.6% positive and which cannot be measured by grep. That requires a reading pass over the examples, and it should be added to the per-chapter review in `REVIEW-CHECKLIST.md` rather than guessed at here.

Nor have I verified that the proposed exercise solutions are feasible for every chapter. Some topics, cloud provisioning and team Git among them, may not admit a self-checkable exercise on a shipped dataset, and those chapters may have to remain open-ended with the limitation stated to the reader.

Finally, the effort estimates are estimates. No part of this plan has been executed, and nothing in it has been tested against the actual difficulty of converting the first display block or writing the first solution set.

On the reliability of the first draft. [**added 17:40 PDT**] The re-measurement that produced this revision found three substantive errors and two omissions in a document that presented its counts as reproducible. The license error is the serious one, because the first draft's remedy happened to be right while its diagnosis was wrong, and a reader acting on the diagnosis alone would have believed the book was merely mislicensed rather than actively non-open. The `11-quarto` figure error came from crediting a chapter with output it does not produce. Both were the result of trusting a scan without spot-checking its output against the files. The counts in this revision were verified against `analysis/report/` directly on 2026-09-07, but they are still grep over source rather than inspection of rendered output, and the same class of error remains possible.

On the two Finding 3 targets specifically. The 170 code-bearing sections is a count of sections containing a fence of any kind, including `bash`, `yaml`, and `sas` blocks that will never render an artifact, so it is an upper bound on what the conditional rule can yield and the achievable number is lower. It will also move once Finding 2's conversion runs. The ambient target is now the April image policy's own five portraits rather than a number derived here, which removes the print-cost question from the critical path; if a larger photographic budget is ever revisited, the claim that color interior printing would defeat a sub-ten-dollar paperback is stated from general knowledge and needs a vendor quote before it is relied on.

Now verified, 2026-09-08. The book was rendered to both HTML and PDF. Both complete clean with no warnings. `toc: true` under `scrbook` produces a correct multi-level table of contents with page numbers, and the trim is the intended 6.5 by 9 inches (468 by 648 points).

And the page estimate was wrong by a factor of nearly three. The book is 712 pages, not ~248. Every draft of this plan repeated the 248 figure, derived by dividing 111,865 words by a conventional 450 words per page. That divisor is for a dense prose page on a larger trim. This book is set at 11pt on a 6.5 by 9 inch page, and a large share of its content is code and rendered output, which occupy a page at a fraction of prose density. The estimate should not have been carried through four revisions without a render to check it, and the lesson is the same one this plan keeps relearning: build the artifact before quoting numbers about it.

Consequences for the plan. Finding 8 argued against restructuring toward Downey's ~140-page target on the grounds that 248 was defensible for the scope. At 712 that argument is much weaker, and length becomes a live question rather than a settled one. Note also that Phase C *added* pages by converting display blocks into executed chunks with output, so this number will keep rising. The print-on-demand item in Phase D needs rethinking: a 712-page paperback is a different product, and a different price, from the sub-ten-dollar edition the evidence base describes. Cultural relevance remains unassessed. The contrast ratios are computed from the hex values in `practicum.scss` rather than measured against rendered output, and no dark-mode variant was examined. Whether the worked examples are in fact worked, as opposed to described, has not been checked by reading them.

Reading pass (2026-10-09)

- **Reading coverage.** The reading was done by six parallel readers with a shared rubric. Cross-group consistency judgments (for example, 'appears in fourteen chapters') are aggregated from their reports and are approximate.

- **Defect verification.** Of the defects in Part 6, six were confirmed directly. The rest were inspected by one reader each, and some rest on that reader's knowledge of tool APIs (renv, Quarto, rticles, ellmer) rather than on execution.
- **Measurement methods.** Reading-time estimates use 200 words per minute including code. The 712-page count comes from Finding 8, not from a new measurement.
- **Source-reading depth.** Several effect sizes come from secondary summaries rather than full papers, notably Rey (2012) and Rowland (2014). The literature review found no controlled evidence on optimal chapter length, glossary tooltips, or webR exercises, so recommendations on those points are inferred.
- **No student testing.** Nothing in this plan has been tested with students. The pilot loop in Part 8 is the mechanism for correcting it.

Expert and LLM-section reviews (2026-10-09)

- **Simulated reviewers.** The three expert reviews were produced by model reviewers instructed to adopt expert personas. They read the source in full and ran checks, and the plan author re-confirmed three of their critical claims against the source (the SAS statement, the dbGaP deposit, and the inner `mice` seed). They are not a substitute for human expert review, which Part 10 requires before the catalog listing.
- **Coverage.** Each reviewer read only assigned chapters (10, 14, and 19 of them). Some chapters were read by two reviewers and some by one; no chapter was read by all three.
- **Evidence status in Part 6.** Items labeled 'E' rest on the reviewer's expert knowledge (regulatory item numbers, policy dates, vendor APIs) and must be checked against primary documents during editing. Reviewers disagreed in detail on one verified result (two versus five distinct datasets in the reseeded simulation); both establish that the simulation is invalid.
- **Currency of the LLM review.** Statements about current tools and practice reflect knowledge as of mid-2026 and will date fastest of anything in this plan. Product names in 4.6 belong in the dated 'Tooling as of' appendix, not in chapter prose.

Appendix. Reproducing the measurements

The two scan scripts used for the first draft are in the session scratchpad: `measure.sh` (per-chapter words, exercises, figures, chunks, retrieval sections) and `measure2.sh` (chunk types, alt text, formats, license, exercise style). They are read-only over `analysis/report/` and can be rerun after each phase to confirm the acceptance criteria above.

The corrections in this revision came from a second, independent scan on 2026-09-07 at 17:40 PDT, scripted as `chk1.sh` through `chk9.sh` in the session scratchpad. The checks that found the errors are worth stating in full, since they should be part of any future verification:

- `grep -c '^```{r' *.qmd` for executable chunks, per file, which is what showed `11-quarto` and `12-rmd-workflow` at zero.
- A loop reporting files where both the mermaid count and the `{r}` count are zero, which is what produced the five-file zero-figure list.
- `grep -i '^license' DESCRIPTION`, `head -8 LICENSE.md`, `grep -i license CITATION.cff`, and the License section of `index.qmd`, read as four separate files rather than assumed to agree.
- `grep -n '^#\+ ' <chapter>.qmd` to produce a heading map, which is what revealed that `Prerequisites` `answers` is always the last heading.
- `grep -rn 'toc' _quarto.yml`, which is what found `toc: false` under the `pdf` format.

The general lesson is that a count aggregated across files hides the per-file zeros, and the per-file zeros are where the defects are.

The section counts underlying the two Finding 3 targets come from `sec.sh` (technical sections per file, defined as `^##` headings less the eight boilerplate headings) and `secode.sh` with `secode.awk` (of those, how many contain a code fence). They yield 374 technical sections, 170 code-bearing and 204 prose-only, and the part count of 8

comes from `grep -c '^    - part:' _quarto.yml`. Rerunning these after Phase C converts display blocks to chunks will move the code-bearing count, so re-measure rather than treating 170 as fixed.

Principal sources

- Dunlosky, J., et al. (2013). Improving students' learning with effective learning techniques. https://www.psychologicalscience.org/news/releases/which-study-strategies-make-the-grade.html
- Adesope, O., et al. (2017). Rethinking the use of tests. https://education.wsu.edu/documents/2018/01/rethinking-use-tests.pdf/
- Cepeda, N. J., et al. (2006). Distributed practice in verbal recall tasks. https://pubmed.ncbi.nlm.nih.gov/16719566/
- Brunmair, M., and Richter, T. (2019). Similarity matters: a meta-analysis of interleaved learning.
- Ginns, P., Martin, A. J., and Marsh, H. W. (2013). Designing instructional text in a conversational style: a meta-analysis.
- Kirschner, P., Sweller, J., and Clark, R. (2006). Why minimal guidance during instruction does not work.
- Wilson, G. *Teaching Tech Together*. https://teachtogether.tech/en/
- The Carpentries. Lesson development handbook. https://docs.carpentries.org/handbooks/lesson_developers.html
- Timbers, T., and Cetinkaya-Rundel, M. Ten simple rules for teaching data science. https://arxiv.org/html/2602.02874v1
- Wickham, H., Cetinkaya-Rundel, M., and Grolemund, G. *R for Data Science* (2e), introduction. https://r4ds.hadley.nz/intro.html
- Bryan, J. *Happy Git and GitHub for the useR*. https://happygitwithr.com/
- Nielsen Norman Group. How users read on the web. https://www.nngroup.com/articles/how-users-read-on-the-web/
- Delgado, P., et al. (2018). Don't throw away your printed books. https://www.uv.es/lasalgon/papers/Delgado%202018%2
- Faded Parsons problems. https://acelab.berkeley.edu/projects/faded-parsons
- Diataxis documentation framework. https://ubuntu.com/blog/diataxis-a-new-foundation-for-canonical-documentation

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Rendered on 2026-10-09 at 17:31 PDT. Source: ~/prj/tch/02-practicum/REFACTOR-PLAN.md